

Technical Training

Cooling System

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ACADÉMIE Vast auto

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Training objectives

By the end of this training, you will be able to:

– Explain how the system removes heat and the role of each component
– Choose the right coolant: families, specifications, colour and mixing traps
– Test the coolant and the system, and carry out an overheating diagnosis
– Diagnose thermostats, water pumps and auxiliary pumps
– Understand active thermal management and electric vehicle circuits

What has changed in 10 years

Yesterday

– A belt-driven pump
– A wax thermostat
– A single circuit
– Green antifreeze every 2 years

Today

– Electric and variable pumps
– Controlled thermostats and thermal management modules
– Several loops: intercooler, electronics, battery
– Long-life and low-conductivity coolants

KEY POINT — The system is now diagnosed as much with the scan tool as with the pressure tester.

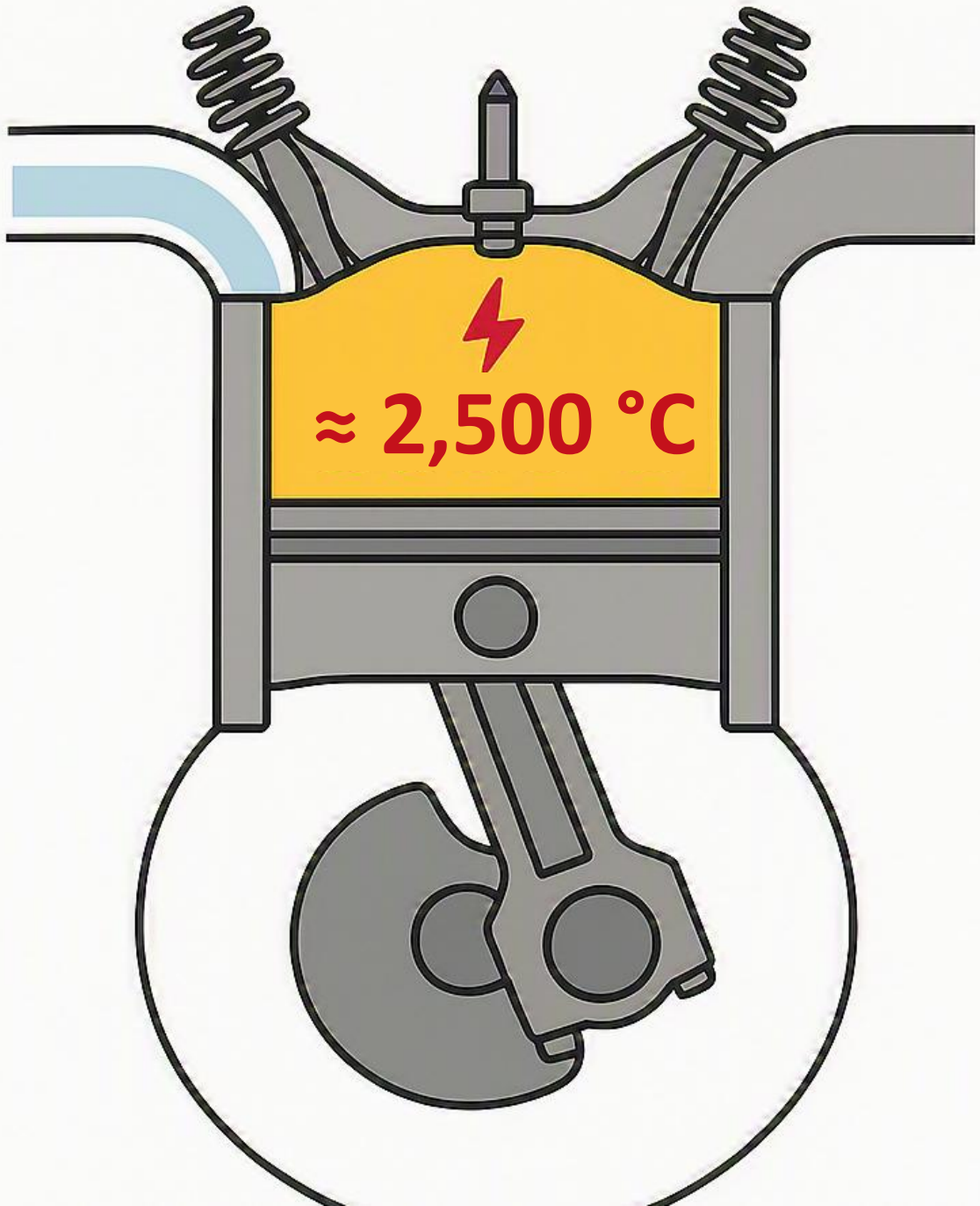
Where does the fuel's energy go?

The balance of a gasoline engine

– About 1/3 moves the vehicle
– About 1/3 leaves through the exhaust
– About 1/3 goes into cooling and losses

KEY POINT — The cooling system removes as much energy as the energy that drives the vehicle.

GOOD TO KNOW — The proportions vary with the engine and the load. The 2,500 °C peak lasts only an instant; in a diesel, it approaches 3,000 °C.

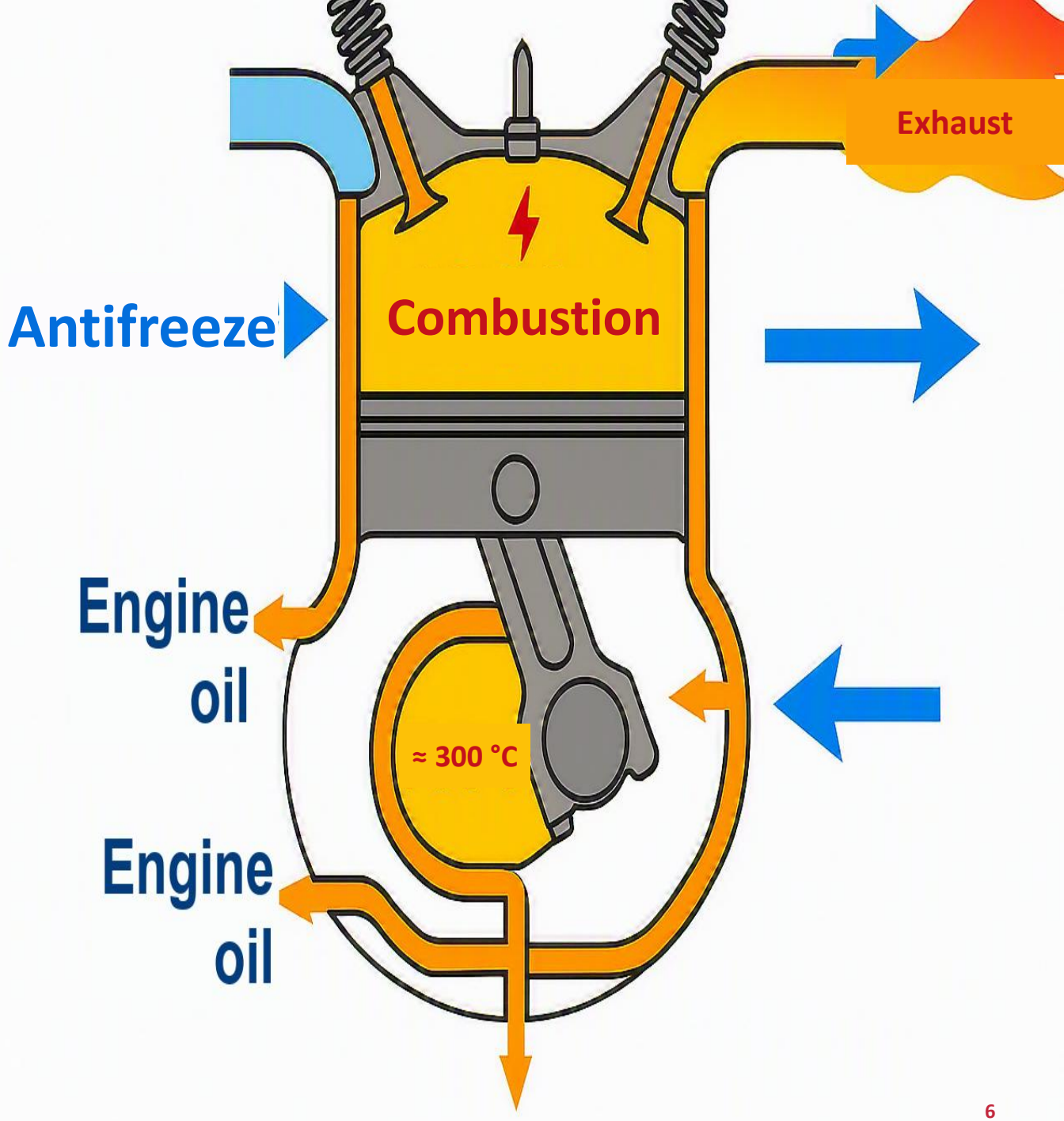


Cooling system

Two fluids carry the heat away

– Antifreeze
– Engine oil

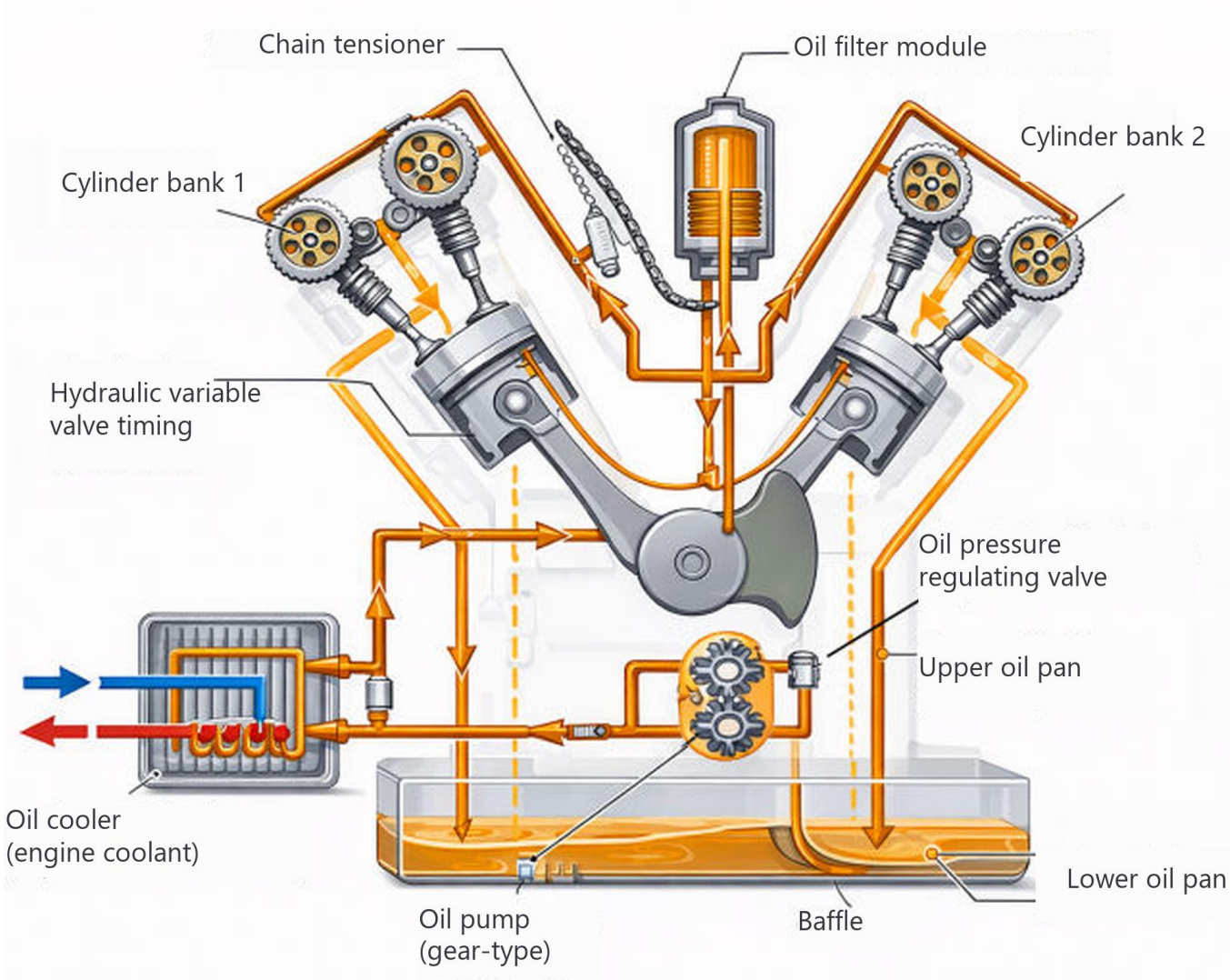
GOOD TO KNOW — The crown of an aluminium piston runs at about 270 to 350 °C. Beyond that, aluminium loses its strength: hence the oil jets.



The role of engine oil

Three roles

– Lubricate, to reduce wear
– Cool, mainly the piston
– Clean



Engine oil

The circuit and its six components

1. Oil pan
– Reservoir at the bottom of the engine, it holds the engine oil

2. Oil pump
– Draws oil from the pan and sends it under pressure throughout the engine

3. Oil filter
– Cleans the oil: particles, metal shavings, contaminants

4. Oil galleries
– Small internal passages in the engine block
– Distribute oil to the crankshaft, the camshaft and the pistons

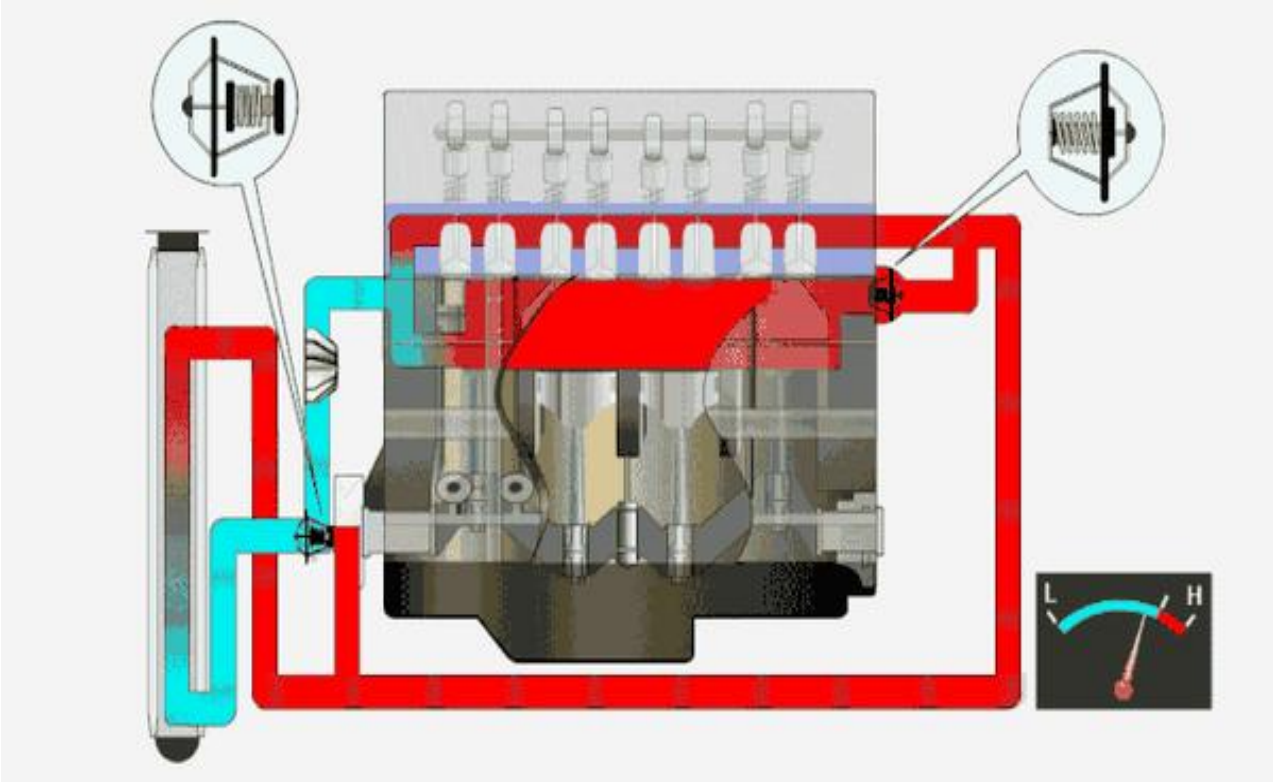
5. Oil jets (piston cooling jets)
– Spray oil under the piston and cool its crown (about 300 °C)

6. Pressure relief valve
– Regulates oil pressure and prevents excess pressure

The role of antifreeze

Two temperature levels to hold

– Keep the engine at a safe temperature, ~90 to 100 °C overall
– Cool the walls, ~300 °C locally right after combustion



Antifreeze

The circuit and its seven components

- **1. Coolant (antifreeze)**
  – A mix of water and anticorrosion and antifreeze additives, it absorbs the engine's heat
- **2. Water pump**
  – Circulates the coolant through the system
- **3. Water jackets**
  – Passages around the cylinder and the cylinder head
- **4. Radiator**
  – Cools the coolant with outside air
- **5. Thermostat**
  – Regulates temperature: closed when cold, open when hot
- **6. Fan**
  – Forces air through the radiator
- **7. Expansion tank**
  – Absorbs the expansion of the coolant

The radiator

An air-to-liquid heat exchanger

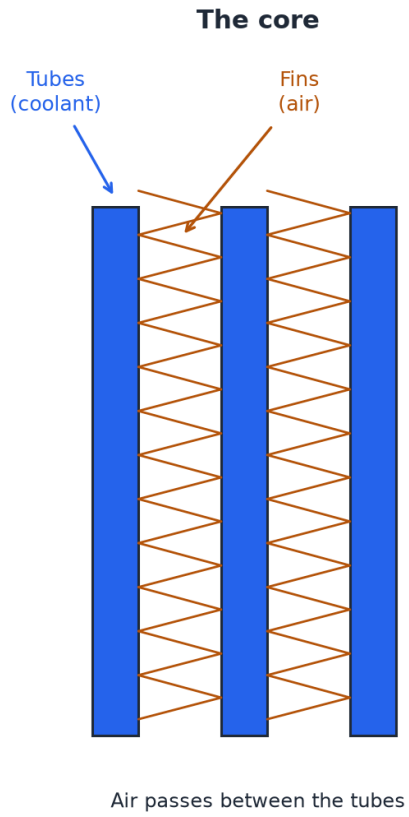
How it is built

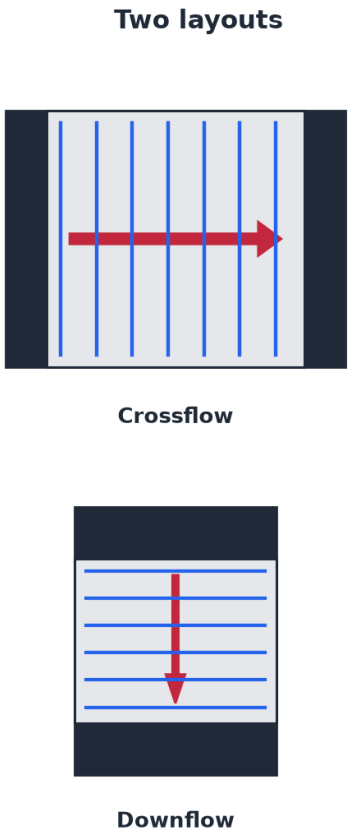
– A core of flat tubes joined by fins, air passes between the tubes
– Tanks at each end, plastic on almost every modern vehicle, crimped onto an aluminum body
– Two layouts: crossflow, tanks on the sides, or downflow, tanks on top and bottom

What puts it out of service

– Internal blockage: deposits, sludge, scale from tap water
– External blockage: insects, leaves, road salt, crushed fins
– Leak at the plastic-to-aluminum crimp, the weak point of the modern radiator

WARNING — Never bolt a radiator to its support without leaving room for thermal expansion. The rubber isolators are there for that.





Conventional or reverse flow

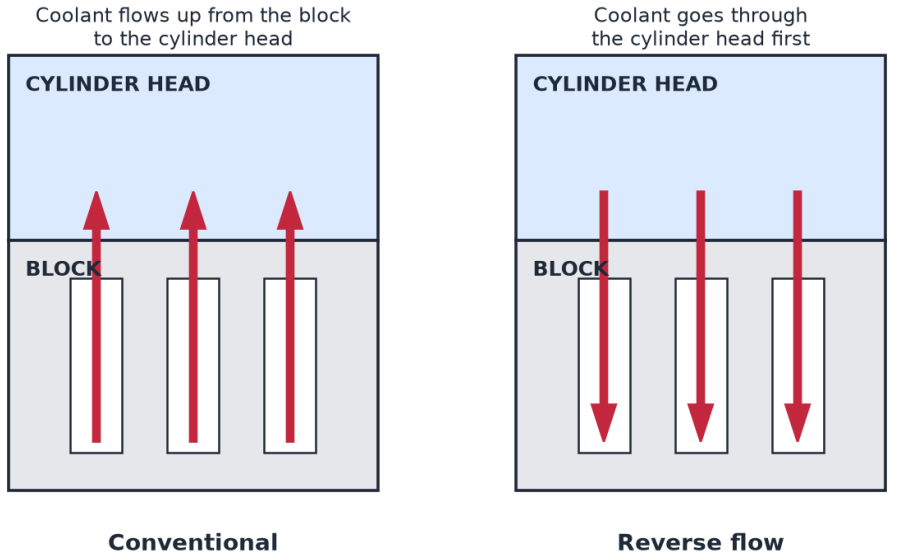
Two ways to circulate the coolant

The conventional system

– The coolant flows upward, from the block to the cylinder head
– It overcools the cylinders in order to cool the heads enough
– The upward flow keeps vapor from being trapped in the heads

The reverse-flow system

– More efficient cooling, better engine efficiency
– Vapor bubbles form against the direction of flow and collect in the heads
– Venting becomes critical, and is handled by small bleed vents



The pressure cap

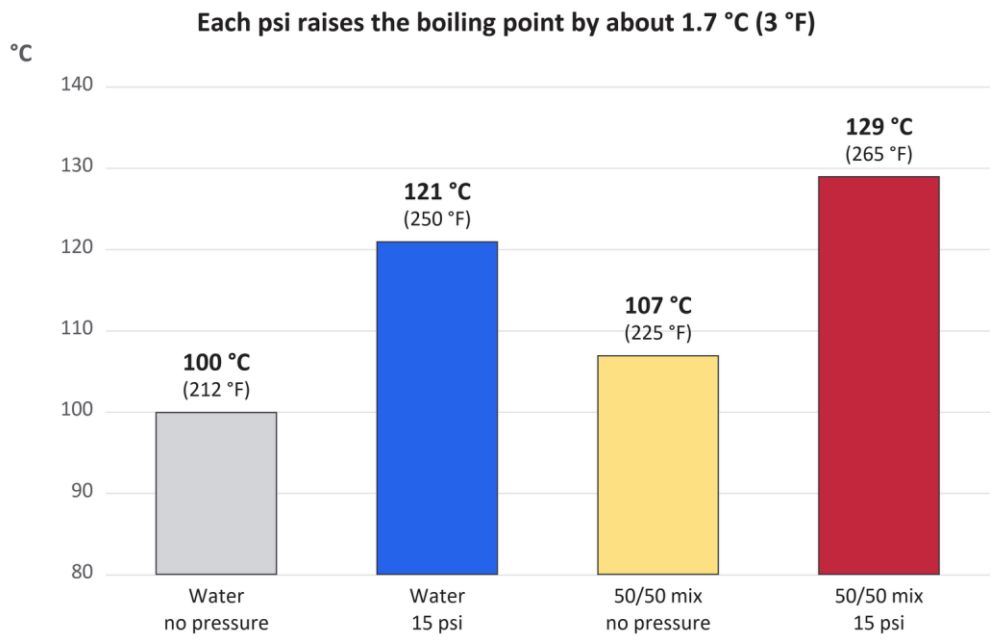
The boiling point multiplier

What it does

– It pressurizes the system, most systems run around 15 psi
– Each psi raises the boiling point by about 1.7 °C (3 °F), a rough rule
– A 50/50 mix boils at about 107 °C (225 °F) in open air, and at about 129 °C (265 °F) under 15 psi

Why it matters

– Vapor does not absorb heat. As soon as local boiling starts, heat transfer stops at that spot
– A cap that no longer holds its pressure will boil a system that is perfectly full



KEY POINT — A cap costs a few dollars and takes thirty seconds to test. It is the most overlooked part of the system.

GOOD TO KNOW — The “3 °F per psi” rule is an approximation: at 15 psi it gives 125 °C for water, while steam tables give 121 °C.

The expansion tank

Recovery reservoir or pressurized tank

Two layouts, two behaviors

– The recovery reservoir: the cap is on the radiator, the reservoir is not pressurized, it takes the overflow and gives it back to the cooling system
– The pressurized tank: the cap is on the tank, there is no cap on the radiator, the whole system is pressurized up to the tank

Reading the level correctly

– The level is read with the engine cold, at the MIN and MAX marks
– A level that rises when hot and drops when cold means the system is breathing properly
– A level that rises and never comes back down, or continuous bubbling, points to combustion gases

The fan and the airflow

It is not an accessory

The three families

– The mechanical fan with a viscous clutch, belt driven, which slips more or less depending on the temperature of the air coming through the radiator
– The conventional electric fan, one or two speeds, controlled by relay and resistor
– The PWM-controlled or brushless electric fan, variable speed, controlled by a module

What gets forgotten

– Air deflectors, baffles and shrouds are not options
– Without them, air goes around the radiator instead of through it





Diagnosing the fan

Three checks before condemning the part

The viscous clutch, engine off and cold

– It must offer some resistance when turned by hand
– If it spins freely more than 1 to 1.5 turns, it slips too much, replace it
– If it is locked or very stiff, it has seized, replace it as well
– Oil streaks radiating from the hub mean the silicone is leaking

Play

– Move the blades by hand: any play comes from a bearing, either the clutch bearing or the water pump shaft bearing

The electric fan

– Command it with the scan tool before reaching for the multimeter
– A fan running at 100 % all the time with a communication code is the fail-safe mode, not a fan failure

The controlled electric fan

The BMW example: the electronics are in the fan

How it is controlled

– The DME powers it through a relay and sends it a 100 Hz PWM signal
– The fan's electronics turn that signal into a speed
– Speed follows the coolant temperature at the radiator outlet and the A/C pressure

Reading the signal

– 7%: standby; 11%: minimum speed, one third of rated speed; 93%: maximum speed
– 97%: fan self-diagnosis command
– Below 7% and above 93%: the signal is used to detect faults, not to run the fan

GOOD TO KNOW — A fan that runs up to 11 minutes after the engine is shut off is the after-run, not a stuck relay.

The heater core

The circuit that runs through the cabin

What it is

– A small radiator placed in the HVAC housing, fed with hot coolant from the engine
– Coolant flows through it constantly on most vehicles, it is the blend door that sets the temperature

No heat, four causes in order

– Low level or air pocket, the number one cause after a service
– Thermostat stuck open, the engine never reaches its temperature
– Plugged heater core
– Faulty blend door or its actuator, on the HVAC side

TIP — Feel both heater hoses by hand. Hot in and warm out is normal. Hot and cold means it is plugged.

The radiator is no longer alone

Everything that draws heat from the same airflow

What stacks up in front of the radiator

– The A/C condenser
– The turbocharger charge air cooler
– The transmission oil cooler
– The engine oil cooler
– The power steering cooler



What that changes

– Every added heat exchanger warms the air reaching the radiator
– A clogged condenser will overheat an engine whose cooling system is perfectly healthy

Active grille shutters

A door in front of the radiator

What they do

– Motorized shutters in front of the radiator, controlled by the engine computer
– Closed: less drag, the engine and the cabin warm up faster
– Open when the coolant is hot or when the A/C needs it

The real case: GM bulletin 16-NA-203

– Chevrolet Volt 2016-2017: check engine light in extreme cold, code P059F (shutter 1 performance)
– Causes: harness chafed near the actuator connector, ice or snow in the shutters
– GM: do not replace the shutters because of ice

GOOD TO KNOW — Shutters stuck closed: the engine runs hot under load or in summer. Before looking further, check that they move.

The transmission cooler

Inside the radiator, and that is where it gets complicated

How it is built in

– On many vehicles, the transmission cooler is a coil housed inside the radiator tank

The trap

– A breached wall between the two lets coolant into the transmission, or the other way around
– The transmission fluid turns pink and milky

What nobody sees coming

– A bad cooler destroys a transmission six to twelve months later, and nobody connects it to the radiator by then



The liquid-cooled intercooler

The turbo joins the cooling system

What it is

– On many recent boosted engines, the intake air is cooled by an air-to-liquid exchanger instead of an air-to-air one
– That exchanger often has its own low-temperature circuit, with its own electric pump and its own small radiator

What it means in the shop

– One more circuit to bleed, with its own procedure
– One more electric pump that can fail, with symptoms of power loss rather than overheating

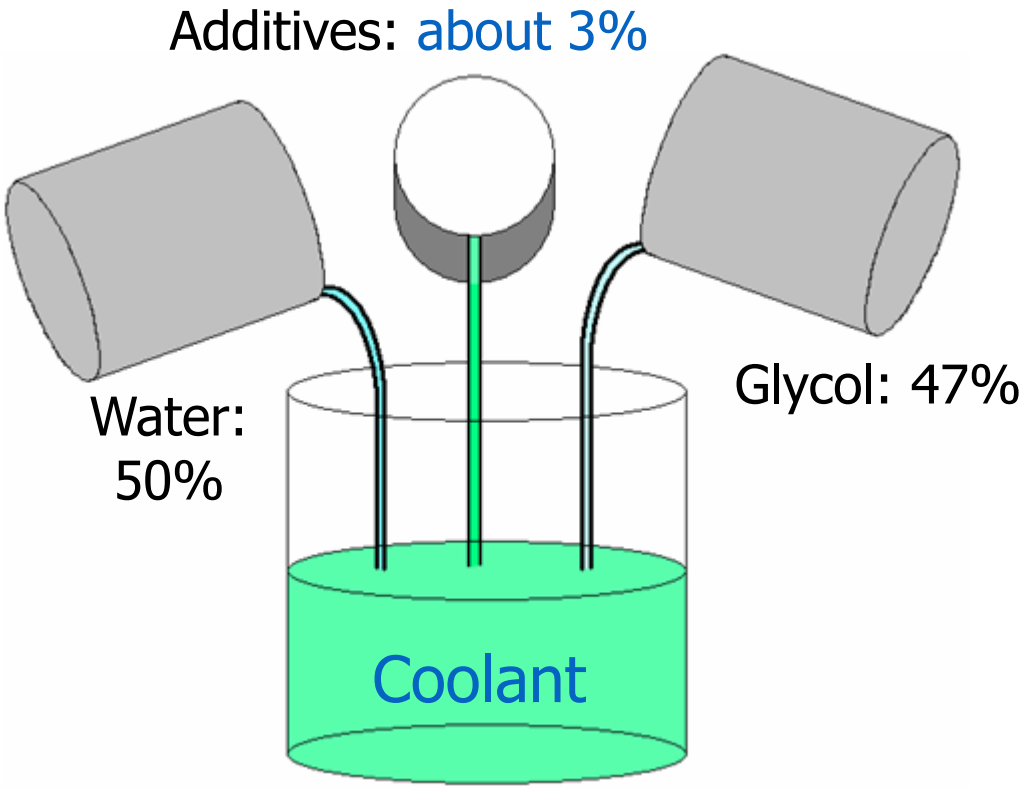
What is coolant?

Three ingredients

– Water
– Glycol, the antifreeze
– Additives

Two ways to buy it

– Concentrate: about 95% glycol, 1.5 to 4% inhibitors, a little water, an antifoam and a dye
– 50/50 premix: about 50% water, 47% glycol and 3% additives



GOOD TO KNOW — The two figures do not contradict each other: 95% describes the bottle of concentrate, 47% what circulates in the engine.

KEY POINT — Concentrate never goes into a system straight.

What we ask of the coolant

Manage temperature

– Carry heat
– Not freeze
– Not boil

Protect the system

– Against corrosion
– Against cavitation
– Against foaming
– By keeping a stable pH

KEY POINT — No single ingredient fills these seven roles: the mix covers them.

Water or glycol?

	Water	Pure glycol	50/50 mix
Heat transfer	Excellent	Poor	Good
Freezes at	0 °C	−13 °C	−37 °C
Boils at, no pressure	100 °C	197 °C	107 °C
Boils at, 15 psi cap	121 °C	—	129 °C

GOOD TO KNOW — The cap gains about 25 °C. A cap that no longer holds pressure means coolant that boils sooner.

KEY POINT — Each covers the other's weakness: water carries heat, glycol protects against freezing and boiling.

Why 50/50, and not stronger?

More glycol does not mean more protection

– At 50%, the mix freezes at −37 °C; at 60%, at −53 °C
– Pure glycol freezes at −13 °C: worse than the mix
– The more glycol, the less heat the coolant carries

The rule

– 50/50 everywhere
– Up to 60% only if the manufacturer allows it for extreme cold

GOOD TO KNOW — The trap: the customer who “strengthens” the antifreeze with straight concentrate. Beyond 60 to 70%, protection goes down instead of up.

WARNING — An over-concentrated system runs hotter in summer.

The water we put in

Never tap water

– Calcium and magnesium form scale with the additives
– Chlorides and sulfates cause corrosion

The right water

– Combustion engine: distilled or demineralized
– Electric vehicle: deionized

GOOD TO KNOW — On an EV, tap water adds ions: it makes the coolant conductive, right next to the high voltage.

WARNING — A low-conductivity coolant is never diluted, not even with deionized water.

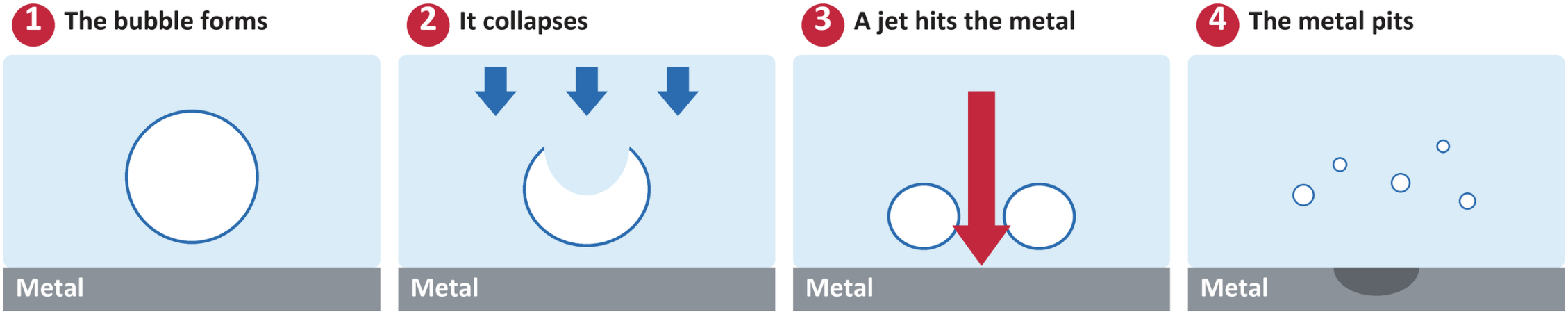




Cavitation: what happens

The mechanism

– Where pressure drops (pump inlet, back of the vanes, wall of a vibrating liner), the coolant forms vapour bubbles, even when it is not hot
– When pressure rises again, the bubble collapses against the metal: a micro-jet of liquid strikes the surface
– Repeated millions of times, it digs holes



GOOD TO KNOW — Like water boiling at altitude: if pressure drops enough, the liquid vaporizes without heating. On a diesel, it is the vibrating liner that creates the low pressure.

Cavitation: where it strikes

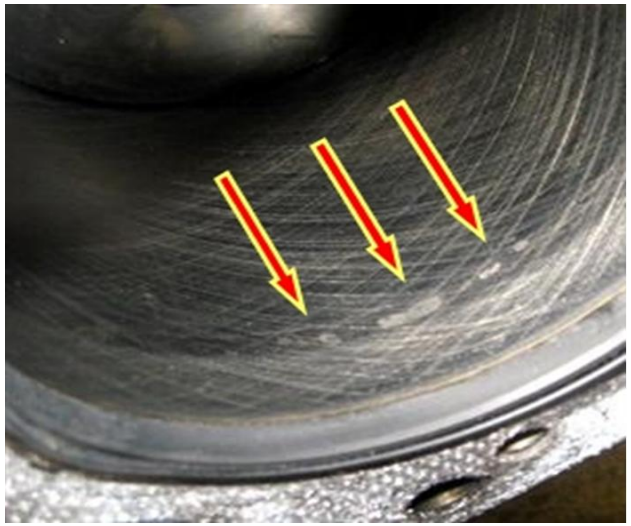
The areas affected

– Diesel cylinder liners, the classic case
– The impeller and housing of the water pump, on gasoline engines too
– Aluminium, much more sensitive than cast iron
– Some engines and some pumps, more prone than others

GOOD TO KNOW — On a heavy-duty diesel, that is why the coolant contains nitrites, topped up with additives (SCA).



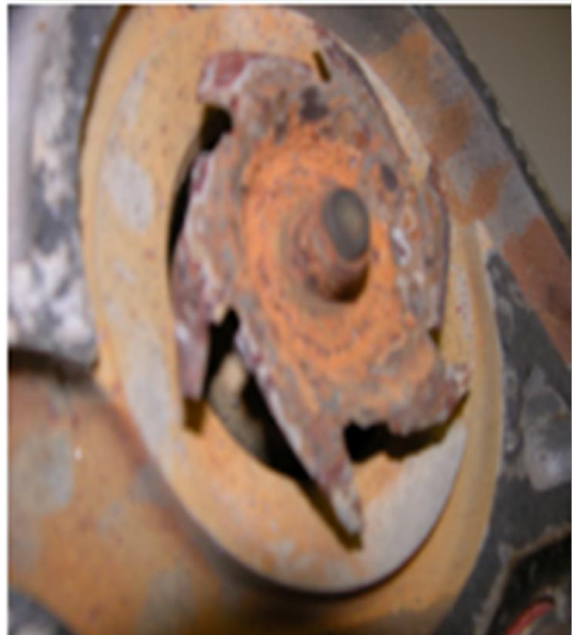




Cavitation: what triggers it

The causes

– Coolant too diluted, or the wrong coolant
– A weak cap: less pressure, bubbles form more easily
– A restriction at the pump inlet
– A low-quality pump
– Extreme conditions





GOOD TO KNOW — Faced with an eaten-away pump, replacing it is not enough: check the coolant type, its concentration and the cap.

WARNING — Cavitation can destroy a water pump in as little as 60 days.

KEY POINT — An eaten-away pump is often the first sign of contaminated or wrongly chosen coolant.

Additives: who does what

They are chosen according to

– Water quality
– The pump design
– The metals
– Plastics
– The cylinder liners
– The service interval

The five main inhibitors

– Silicates
– Phosphates
– Borates
– Nitrites
– Organic acids

GOOD TO KNOW — The glycol is the same in almost every coolant: what changes from one family to the next is the 3% of additives.

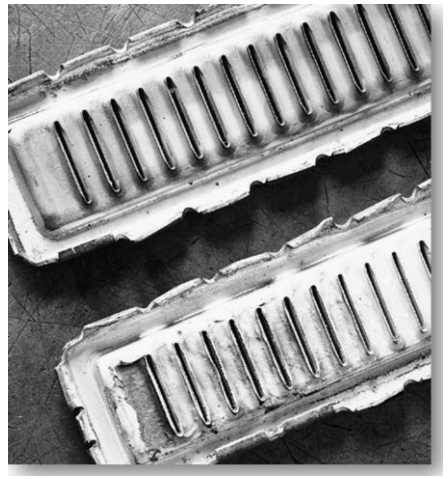
KEY POINT — It is the choice of additives, not the colour, that sets one coolant apart from another.

Silicates

Their strengths

– The fastest protection on the market
– A protective layer forms on aluminium as soon as the system is filled
– Very good against pump cavitation and abrasion





Their downsides

– They deplete quickly
– At end of life, they form gels that hinder heat exchange, and they can become abrasive
– Deposits on pump seals make the pump weep

GOOD TO KNOW — North American pumps, prone to cavitation, needed this immediate protection.

KEY POINT — Properly dosed, they last a long time. Modern coolants contain little.

Phosphates and borates

Phosphates

– They quickly protect cast iron and steel
– With hard water (calcium, magnesium), they form scale on hot surfaces: hence phosphate-free European coolants

Borates

– They stabilize the pH
– They can corrode aluminium
– Some manufacturers avoid them: Toyota Super Long Life contains none



GOOD TO KNOW — In Japan, water is soft: Japanese coolants keep their phosphates. We come back to this with the three continents.

Nitrites and amines: the “NA” in NAP

Nitrites

– They protect diesel liners against cavitation
– They deplete and are topped up with an additive (SCA)
– They are checked with test strips on diesels

The problem

– A nitrite and an amine form nitrosamines, which are carcinogenic
– Sweden and Norway banned these combinations: hence European NAP coolants, free of nitrites, amines and phosphates
– NAPS: also silicate-free

GOOD TO KNOW — Heavy-duty diesel coolants still contain nitrites: their concentration is checked with strips and topped up with SCA.

Organic acids and 2-EHA

Organic acids (OAT)

– They act mainly where corrosion starts, so they wear slowly: 5 years and more
– Their pH is lower
– In return, their protection is slow to build up

2-EHA

– An organic acid that attacks nylon 6.6 and silicone
– Involved in the GM intake gasket case, covered later

GOOD TO KNOW — Silicates coat everything from the start and get used up; organic acids stay in reserve and act where corrosion starts.

Three continents, three chemistries

North America

– The problem: pumps prone to cavitation
– The answer: silicates

Europe

– The problem: very hard water and a lot of aluminium
– The answer: no phosphates, but silicates

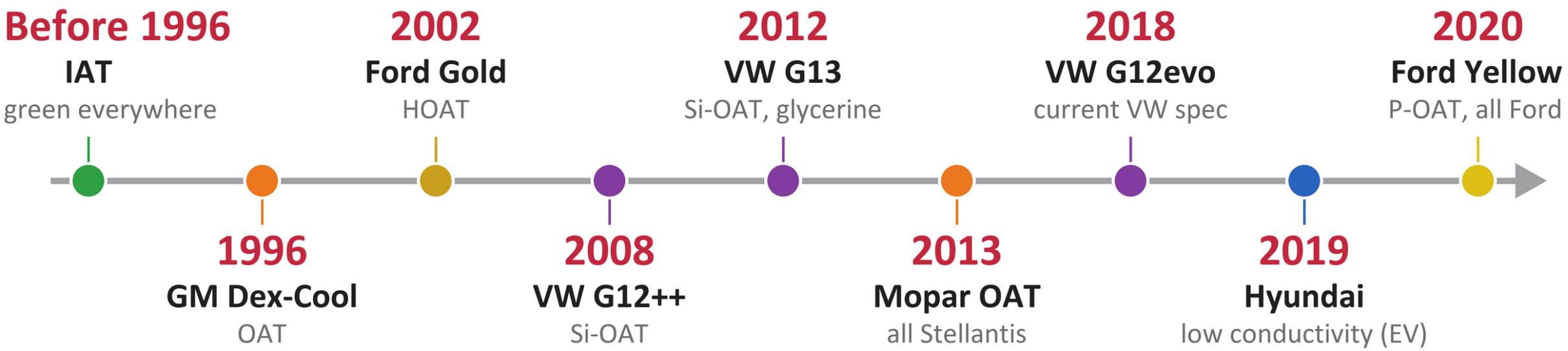
Japan

– The problem: pump seals that do not tolerate silicates
– The advantage: soft water
– The answer: phosphates, no silicates

GOOD TO KNOW — In the 1980s, each market chose its chemistry according to its engines and its water. This map explains all the families that follow.

KEY POINT — That is why there is no universal coolant.

40 years of coolants on one line



GOOD TO KNOW — Each dot has the most common colour of its family: green IAT, orange OAT, gold HOAT, violet Si-OAT, blue low conductivity, yellow P-OAT.

IAT, yesterday's green

What it contains

– Silicates
– Phosphates
– Borates

Its maintenance

– pH of 9 to 11 when new, change at 8.5
– About 2 years
– Ford used it until 2002



GOOD TO KNOW — Mineral inhibitors: very fast protection, but it is used up quickly.

WARNING — IAT is old technology: it can damage ceramic-phenolic seals and recent water pumps.

OAT: the orange of GM and Stellantis

The coolants

– GM Dex-Cool (GM 6277M), since 1996: 5 years or 160,000 km
– Mopar OAT (MS-12106), on all Stellantis vehicles since 2013: 10 years or 240,000 km
– Ford Orange (WSS-M97B44-D), from about 2011 to 2019



Its limits

– Sensitive to dilution and air
– Protection slow to build up

GOOD TO KNOW — At Ford, Orange can be topped up or replaced with Yellow. Mopar OAT replaces the HOAT Chrysler used before 2013.

Dex-Cool: what happened



Cap fouled with sludge



Attacked intake gasket



Sludge at the filler neck

- Intake gaskets (nylon and silicone) on 3.1 and 3.4 L V6 engines, from 1995 to 2003
- Sludge on the 4.3 L
- A class action: GM reimbursed part of the repairs

GOOD TO KNOW — Two problems: 2-EHA attacking the intake gaskets, and sludge forming when air gets in.

WARNING — Low level + air = flush, even if the coolant is recent.

HOAT: the European, and the old Ford and Chrysler

Hybrid Organic Acid Technology

– Silicates and organic acids, no phosphate
– Ford Gold, from 2002 to about 2010; Chrysler before 2013
– Mercedes (MB 325.0) and BMW (LC-87)
– Tesla: G-48, blue







Same family, different colours

GOOD TO KNOW — The “H” means hybrid: organic acids for long life, a little silicate to protect aluminium quickly.

P-OAT: the Asian brands, and Ford today

Phosphates and organic acids, no silicate

– Toyota: pink (Super Long Life)
– Honda: Type 2, blue
– Nissan: blue
– Mazda: FL-22, green
– Hyundai and Kia: pink, since about 2020
– Ford: Yellow, on all Ford vehicles since 2020



GOOD TO KNOW — The “P” means phosphated: no silicate for Japanese pump seals, phosphates because the water is soft.

Si-OAT: the most recent

At VW

– G12++ (TL 774-G) in 2008
– G13 with glycerine (TL 774-J) in 2012
– G12evo (TL 774-L) in 2018: the current spec, mixable with G12, G12+, G12++ and G13

Watch out for the Gs

– VW's Gs are specifications; Glysantin's Gs (G30, G40, G48, G64) are BASF products
– Equivalents: G48 = G11, G30 = G12+, G40 = G12++, G64 = G12evo

GOOD TO KNOW — Organic acids with a small dose of silicate, violet in most cases. G13 replaces part of the glycol with glycerine.

The families at a glance

Family	Key additives	Typical life	Examples	Common colour
IAT	Silicates, phosphates, borates	About 2 years	Older GM, Ford before 2002	Green
OAT	Organic acids	5 to 10 years	GM Dex-Cool, Stellantis 2013 and later	Orange
HOAT	Organic acids + silicates, no phosphate	Depends on the manufacturer	Mercedes, BMW, Tesla, old Ford Gold	Yellow, blue
P-OAT	Organic acids + phosphates, no silicate	Depends on the manufacturer	Toyota, Honda, Nissan, Mazda, Hyundai, Kia, Ford 2020 and later	Pink, blue, green, yellow
Si-OAT	Organic acids + a little silicate	Depends on the manufacturer	VW and Audi (G12++, G13, G12evo)	Violet
Low conductivity	Very few ionic additives	2 to 3 years	Battery of some Hyundai and Kia models	Blue

GOOD TO KNOW — The colour column is a guide only: the same colour exists in several families.

Why an electric vehicle changes everything

What changes

– The coolant runs right against the high voltage: battery, inverter, motor
– Target temperatures are lower: the battery likes 15 to 35 °C
– The metals and fittings are different
– The coolant affects fast charging and battery health

Three strategies

– Conventional indirect: most of today's EVs
– Low-conductivity indirect: mainly Asian brands, should become widespread within 3 to 5 years
– Immersion in a dielectric fluid: a niche (racing, luxury)

GOOD TO KNOW — The coolant runs a few millimetres from the high voltage: if it leaks into a housing, its conductivity becomes a safety issue.

Conductivity, the new criterion

– Conventional coolant: about 3,000 µS/cm; low conductivity: less than 100 µS/cm
– It is the corrosion inhibitors that make the coolant conductive
– Less conductivity = harder protection and shorter life

GB 29743.2 test	Coolant	Result
800 V	Conventional, 3,200 µS/cm	Smoke in 34 min, combustion in 45 min
400 V	Conventional, 3,200 µS/cm	Smoke in 50 min, combustion in 2 h 30
800 V	500 µS/cm	Arcing and combustion in 13 h
800 V	300 µS/cm	No reaction in 72 h

GOOD TO KNOW — The paradox: what protects the metal is also what conducts current.

KEY POINT — New standards are converging on 100 µS/cm for new coolant.

A real case: Hyundai and Genesis

Vehicle	Coolant
Ioniq Electric, Kona EV	Green, then blue low conductivity (2019 to 2021)
Ioniq 5, Electrified G80	Pink for the motor, blue for the battery
Ioniq 6, GV60, GV70	Pink everywhere
Tucson and Santa Fe plug-in hybrids	Pink for the engine, blue for the electric motor and the battery
Tucson and Santa Fe hybrids	Changed from blue to pink

Intervals: blue every 60,000 km or 3 years; pink at 200,000 km or 10 years, then every 40,000 km or 2 years

GOOD TO KNOW — The only exception to “colour means nothing”: here, Hyundai uses it to tell its two coolants apart.

WARNING — At Hyundai, identify the coolant by its colour in the reservoir before any top-up.

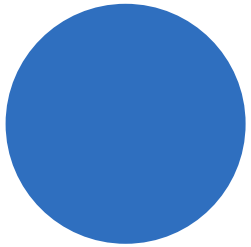
Shop rules on an EV

- Never mix, never dilute, no additives
- The wrong coolant causes a loss of isolation: code P0AA6
- Do not switch from a conventional coolant to a low-conductivity coolant without the manufacturer’s instruction
- Flush with deionized water only
- Bleeding is often done with the scan tool

GOOD TO KNOW — P0AA6: high-voltage isolation fault. Flush water: less than 10 µS/cm (Prestone).

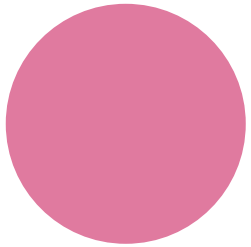
WARNING — Disable the high voltage before any work.

Colour means nothing



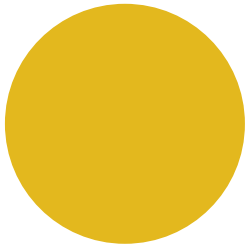
Blue

– G-48 HOAT (Tesla, BMW)
– Honda Type 2, P-OAT
– Hyundai low conductivity



Pink

– Toyota, P-OAT
– Hyundai and Kia, phosphated



Yellow

– Old Ford Gold, HOAT
– Ford Yellow, P-OAT

GOOD TO KNOW — Dye is only dye: it says nothing about the additives. Only the manufacturer's specification counts.

KEY POINT — Choose a coolant by its specification, never by its colour.

Finding the right specification

Where to find it

– The owner's manual
– The service information
– Sometimes the reservoir or the cap

Examples

– GM 6277M
– Stellantis MS-12106
– Ford WSS-M97B57
– VW TL 774-L
– Toyota Super Long Life

"All makes"

– Meets ASTM D3306, but that does not replace the manufacturer's specification
– Even Recochem writes it on its pink EVX coolant: "Do not use in batteries that require a low-conductivity coolant"

GOOD TO KNOW — ASTM D3306 guarantees a proper coolant, not the right additive. An "all makes" coolant: emergency top-up on a combustion engine only, never on a battery.

Mixing two families

What happens

– G11 and G12 mixed together form a brown sludge
– In general, a mix brings the life down to that of the weaker of the two coolants

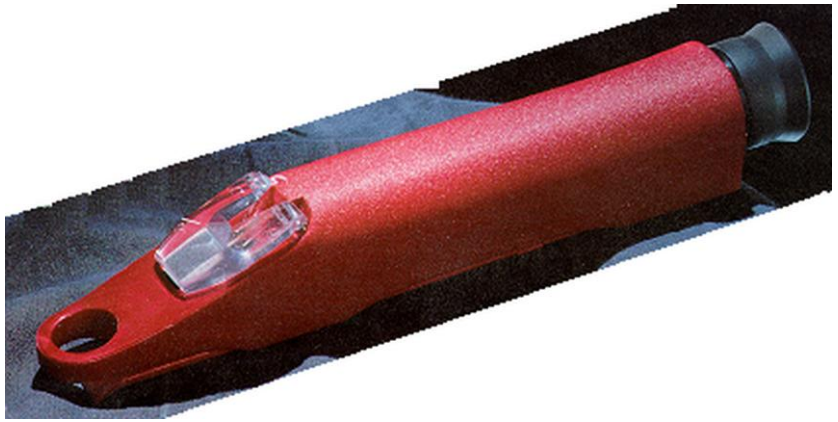
When it is allowed

– Ford Yellow can top up Orange
– G12evo mixes with G12, G12+, G12++ and G13

GOOD TO KNOW — If you do not know what is in the system: complete drain and flush.

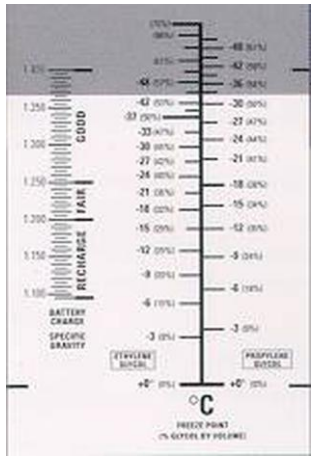
KEY POINT — Mix only when the manufacturer allows it. Otherwise: complete drain.

Measuring freeze protection



The ball hydrometer

– Quick to read, but thrown off as soon as the coolant is hot



Refractometer scale: ethylene on the left, propylene on the right

The refractometer

– More accurate, it compensates for temperature
– Two scales: reading propylene on the ethylene scale overstates the protection

GOOD TO KNOW — This test gives freeze protection, not the condition of the inhibitors: a coolant protected to −40 °C can be chemically dead.

Ethylene glycol or propylene glycol

Ethylene glycol

– The base of almost all factory coolants
– Better heat transfer and better freeze protection
– Toxic and sweet: a danger for children and animals, it is recovered as hazardous waste

Propylene glycol

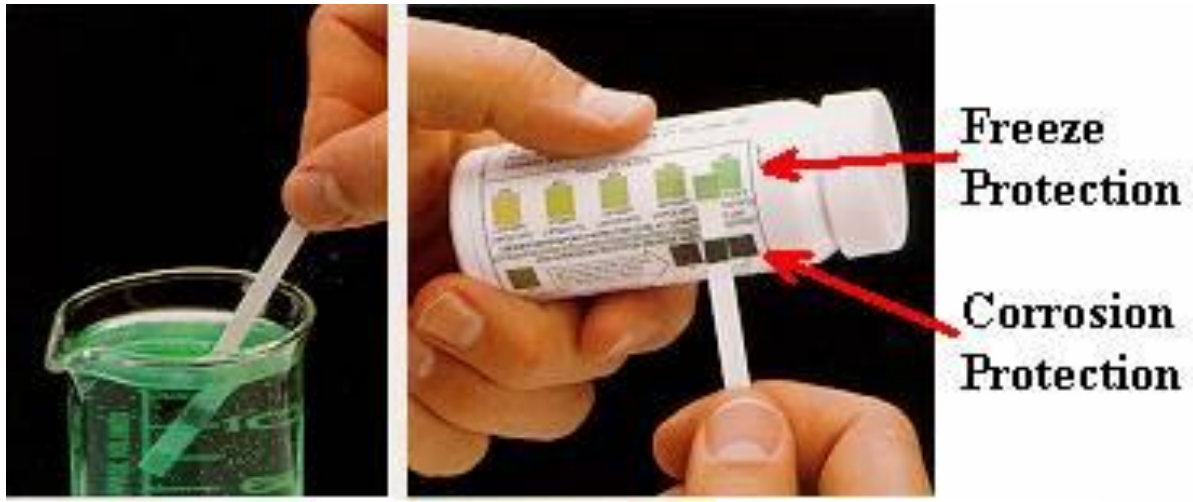
– Low-toxicity aftermarket coolants
– Some specialized heavy-duty diesels
– Do not mix the two: the mix distorts the freeze-point reading

GOOD TO KNOW — Before measuring, know which one is in the system: it decides which scale to read.

pH and test strips

Two ways to measure it

– Test strips: quick, they also give freeze protection and corrosion protection
– The pH meter: more accurate



When to change the coolant

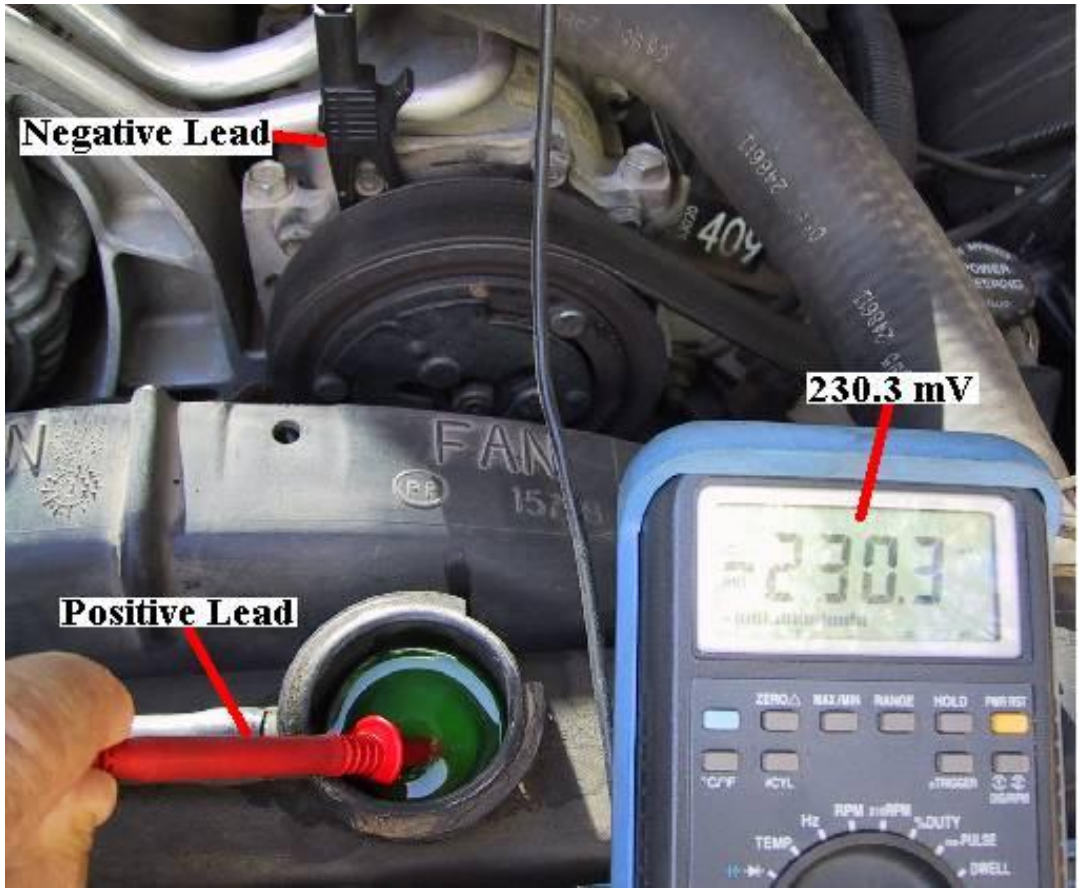
– IAT: below 8.5
– OAT and HOAT: below 7.5





GOOD TO KNOW — pH drops as the inhibitors deplete: it is the best indicator of the coolant's chemical condition.

Coolant voltage



The measurement

– Engine hot and running, all loads on
– Negative lead on engine ground or the negative terminal
– Positive lead in the coolant, without touching metal
– Up to 0.3 V: acceptable. Above: find the faulty ground

GOOD TO KNOW — In the photo, 230 mV: below the limit. Aluminium is more sensitive than cast iron to this current.

Careful: electrolysis

What gets wrongly blamed on it

– Not every unexplained damage is electrolysis
– Pump cavitation and erosion often come from contaminated coolant

Real electrolysis

– A bad ground sends starter or alternator current through the coolant
– It can eat through a heat exchanger in a week
– The remedy: voltage drop tests on the grounds

GOOD TO KNOW — The BMW case covered with thermostats: traces of electrolysis, and BMW has the motor replaced.

The cooling system pressure test

The procedure

– Engine cold, always
– Tester in place of the cap, on the radiator or the reservoir
– Pump up to the pressure marked on the cap, no more (13 to 16 psi in general)
– Leave it under pressure for at least 15 minutes

Reading the result

– The needle must not move
– Drop with no puddle: internal leak (heater core, head gasket) or a leak that evaporates on a hot part
– Drop with a puddle: follow the trace upward

GOOD TO KNOW — Vacuum filling does this test for free: if the vacuum does not hold for two minutes, the system leaks.

The cap test

The procedure

– Fit the cap on the tester adapter
– Pump up to its rated pressure
– It must hold it for several minutes

What we are looking for

– Does not reach its rating: tired seal or spring
– Does not release: the system builds too much pressure
– Hardened, cracked or deformed seal: replace

GOOD TO KNOW — Thirty seconds, on the same tool: a weak cap makes the coolant boil sooner and encourages cavitation.

The combustion gas test

How it works

– Draw the air above the coolant through a blue reagent, without drawing in coolant
– With CO2, blue turns yellow: head gasket, cracked head or block
– Diesel: use the reagent made for diesel

When to think of it

– Coolant loss with no visible leak
– Recurring overheating
– Continuous bubbles in the reservoir, level that rises on its own

GOOD TO KNOW — The reagent must be blue to start with: a reagent that has already turned green gives a false result.

Finding the leak

Traces and tools

– Dry residue: white, green, pink or orange depending on the coolant
– Fluorescent dye and the UV lamp, for slow leaks
– The infrared thermometer, to spot cold areas on the radiator

The most frequent spots

– Plastic-to-aluminium crimp on the radiator
– Plastic quick connectors
– Swollen or cracked hoses
– Thermostat housing

GOOD TO KNOW — A leak running onto a hot part evaporates without leaving a puddle: look for the residue, not the puddle.

Plastic component failure

Plastic has taken the place of metal

– Thermostat housings, thermal modules, radiator tanks, quick connectors
– The most used: nylon 6.6, which handles heat well
– It ages with heat cycles and the coolant chemistry

GOOD TO KNOW — The right coolant also protects the plastics: some organic acids, like 2-EHA, attack nylon 6.6.

The coolant temperature sensor (ECT)

The sensor that drives everything else

What it is

– A negative temperature coefficient thermistor, NTC: the hotter the coolant, the lower its resistance
– The module sends it a 5 V reference voltage and reads the voltage that comes back
– Cold means high resistance means high voltage. Hot means low resistance means low voltage

What it controls

– The fuel calculation and the switch to closed loop
– Fan engagement
– The dash gauge needle, often damped by the module

GOOD TO KNOW — Open wire: voltage rises to 5 V and the module reads very cold. Short to ground: voltage drops to 0 and it reads very hot.

One sensor or two

On modern systems there is rarely just one

Why two

– One sensor at the cylinder head outlet and another at the radiator outlet, or one in each circuit on split systems
– The difference between the two is what lets the module know whether the coolant is actually flowing

In the shop

– Read both values with the scan tool, engine cold: they must match within a few degrees, and equal ambient temperature
– A difference at cold start is a sensor or its wiring, not a flow problem

GOOD TO KNOW — Engine cold since the day before, all temperature sensors should read the same value: it is the quickest test.

Reading the codes the right way

What the code says, and what it does not say

The circuit codes

– P0115 to P0119: sensor circuit, voltage out of range, intermittent fault

The performance codes

– P0125: insufficient temperature to enter closed loop

– P0128: the coolant does not reach thermostat regulating temperature within the expected time

WARNING — P0128 does not mean “faulty thermostat”, it means “the engine does not warm up fast enough”. A sensor reading too low will set it too.

GOOD TO KNOW — Mazda’s P0126 was a backup thermostat opening too early: the code gives the symptom, not the part.

The role of the thermostat

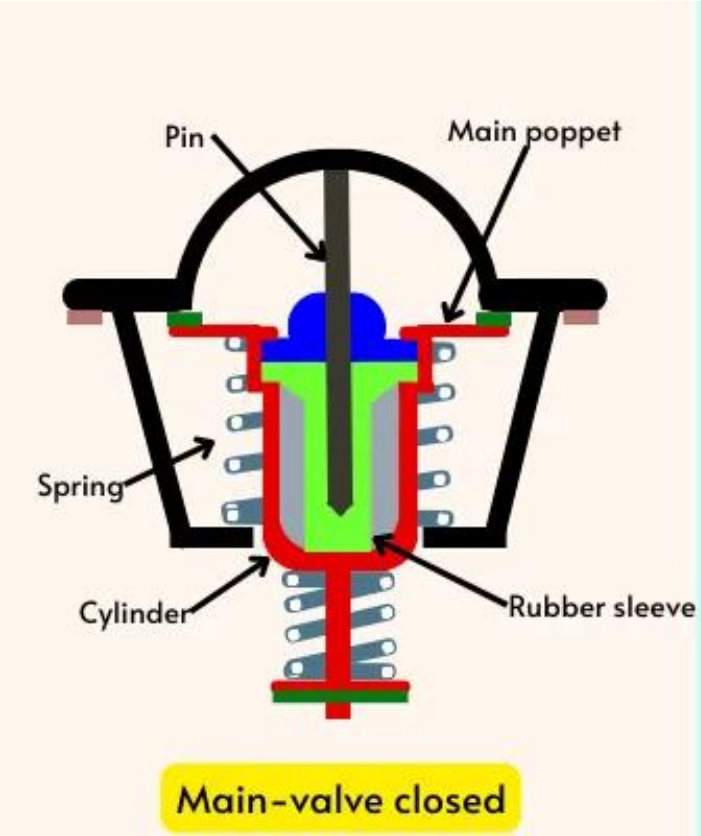
What it does

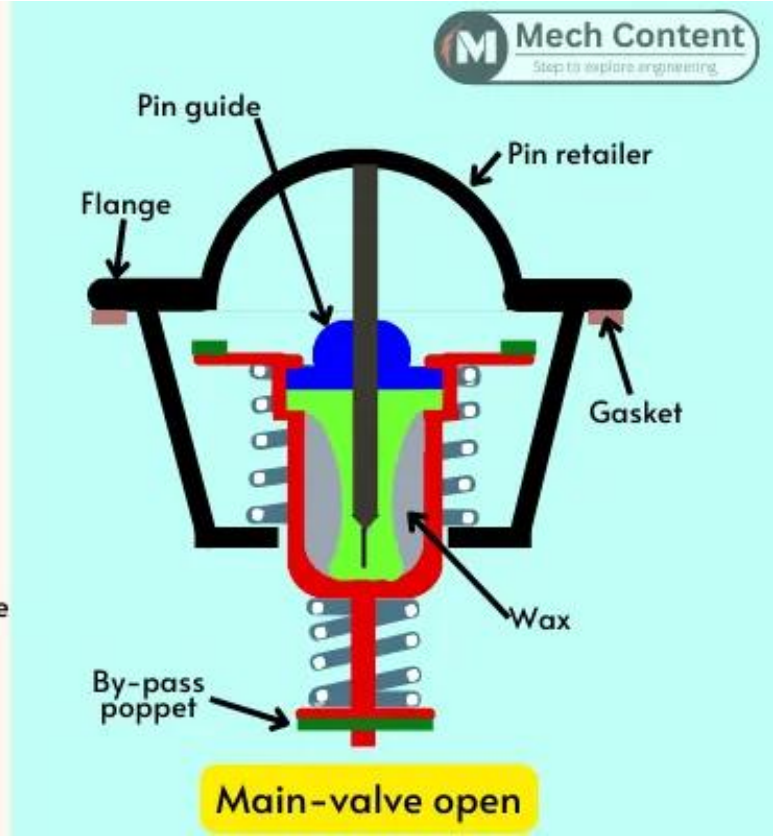
– It regulates engine temperature by metering the flow to the radiator
– Closed when cold: coolant circulates in the engine through the bypass
– Open when hot: coolant goes through the radiator to give up its heat
– It does not create flow: the pump does the pushing



GOOD TO KNOW — A faulty thermostat does not only cause overheating: stuck open, the engine burns more fuel and pollutes more.

Wax thermostat: how it opens





– The wax melts and expands as it heats
– It pushes on the fixed pin: the cylinder moves down
– The main valve opens, the bypass valve closes
– As it cools, the spring closes everything

GOOD TO KNOW — The wax only reacts to the temperature of the coolant touching it: hence the delay, and the benefit of the electronic thermostat.

Wax thermostat: the details that matter

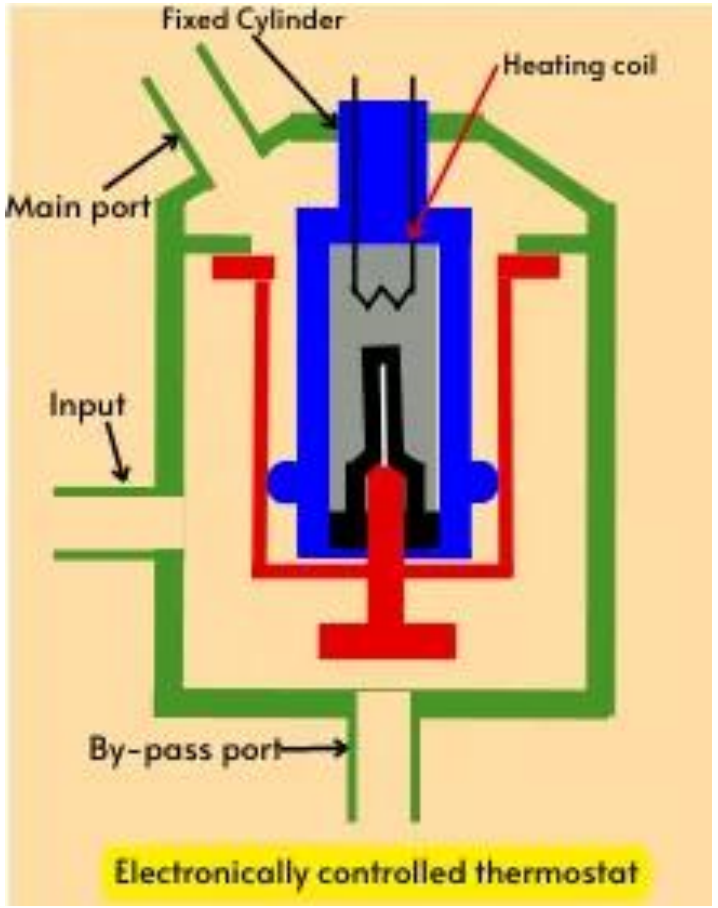
– The gasket: it seals the flange, it is replaced with the thermostat
– The return spring: it keeps the valve closed when cold
– The jiggle pin: it lets air through, during filling and while running; position it according to the manufacturer’s procedure, usually at the top



GOOD TO KNOW — A thermostat installed backwards or a misoriented jiggle pin is a guaranteed air pocket.

Electronic thermostat: how it is built

– A heating element is built into the wax
– The ECM powers it when it wants to open earlier
– With no current, it works like an ordinary wax thermostat
– Also called a map-controlled thermostat: BMW, VW and others



GOOD TO KNOW — It keeps its wax: in case of an electrical fault, it still opens, but later.

Electronic thermostat: what we gain

Wax only

– Load rises, then the engine heats up, then the coolant
– The wax finally expands and the thermostat opens
– Cooling arrives late

With the heating element

– The ECM sees the load rising and heats the wax right away
– The thermostat opens in advance
– Temperature adapts to the driving: hotter in town, cooler at full load

GOOD TO KNOW — An engine running a little hotter in easy driving is not faulty: it is intentional.

The bypass

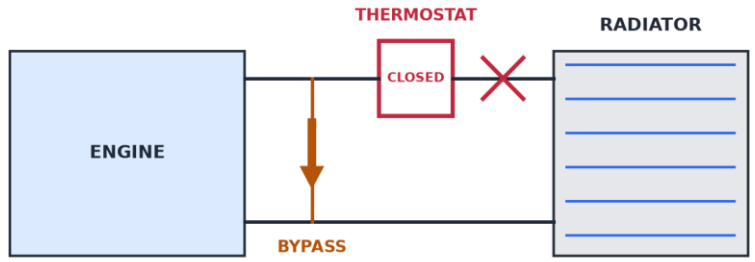
Why the engine does not develop a hot spot with the thermostat closed

What it is

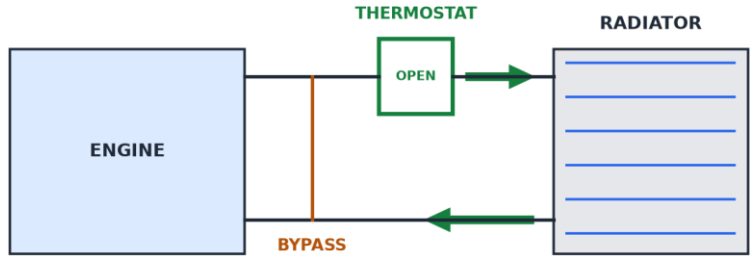
– A passage that lets the coolant circulate through the engine while the thermostat is still closed
– The coolant runs in a short loop, engine only, without going through the radiator

Why it is essential

– Without a bypass, the coolant would sit still in a heating engine: a hot spot at the head and thermal shock when it opens
– This is also the circuit that feeds the heater core, which gives cabin heat before warm-up is complete



The coolant runs in a short loop through the engine and warms up



The coolant goes through the radiator and is cooled

GOOD TO KNOW — The heater gives heat before the engine is warm: it is the bypass that circulates the coolant while the thermostat is closed.

How to read a thermostat

The marking is not the full-open temperature

What the number says

– The marking gives the temperature at which the thermostat starts to open
– It should be fully open about 11 °C (20 °F) higher
– A 90 °C (195 °F) thermostat is therefore fully open around 102 °C (215 °F)



What that changes in diagnosis

– An engine that settles above the marked value is not necessarily faulty, you need to know the full range

GOOD TO KNOW — The number stamped on the part is often in °F: 195 °F is 90 °C.

Testing a wax thermostat

On the bench

– At room temperature, it must be closed
– Heat water to at least 11 °C (20 °F) above its rating
– Immerse the thermostat and follow the temperature on the thermometer
– Let it cool in the water and see if it closes again

Good or bad

– Good: it starts to open within about 2 °C (3 °F) of its rating, it is fully open 11 °C higher, it closes again as it cools
– Bad: open when cold, does not open, does not open fully, or does not close again

GOOD TO KNOW — Never in the microwave. And before removing it: the temperature curve on the scan tool, engine cold to hot, often already tells the whole story.

A real case: BMW and its map-controlled thermostat

BMW bulletin SI B11 18 16 (X1 F48)

– Codes 1D2401, 1D2402, 1D2404: heating element short to positive, to ground, or open circuit
– “Engine overheated” message, coolant sometimes discoloured
– Cause: the wire insulation breaks down and coolant enters the harness

The correction

– Remove the pump and thermostat assembly, look for corrosion and electrolysis
– Nothing: replace the pump, thermostat and bracket assembly
– Traces of electrolysis: BMW has the engine replaced

GOOD TO KNOW — Current leaking into the coolant attacks aluminium: this is the real electrolysis from the slide “Warning: electrolysis”. Same family: recall 21E-A03 on 2022 530i models.

The three failure modes

Open, closed, and the one that gets forgotten

Stuck closed

– Rapid overheating, the most visible and most destructive mode

Stuck open

– The engine never warms up, no cabin heat, P0128
– Higher fuel consumption and higher emissions

Cycling

– The thermostat opens and closes constantly, never settling
– It is the biggest cause of thermal shock in the system

GOOD TO KNOW — A gauge that rises and falls suggests oscillation, but a low level or air gives the same symptom: check the level first.

What thermal shock destroys

A subject documented by SAE

What cycling causes

– Repeated stroke cycles of the thermostat, beyond what it is designed for
– Fatigue of the thermostat flange
– Thermal shock in the radiator
– The weakest point gives out first: the radiator tube-to-header joint

What to take away

– A radiator leaking at the header shortly after a thermostat replacement is not a coincidence

GOOD TO KNOW — After replacing a thermostat that was oscillating, inspect the radiator: the damage is often already done.

The water pump

What it does

– It circulates the coolant between the radiator and the engine
– The impeller, driven in rotation, creates a centrifugal force that pushes the coolant through the circuit

The coolant's path

– It flows through the engine block and the cylinder head to absorb the heat produced by combustion
– It returns to the radiator, where that heat is released thanks to the radiator and the fan

KEY POINT — This continuous cycle keeps the engine at its optimal operating temperature, improves its performance and prevents damage caused by overheating.

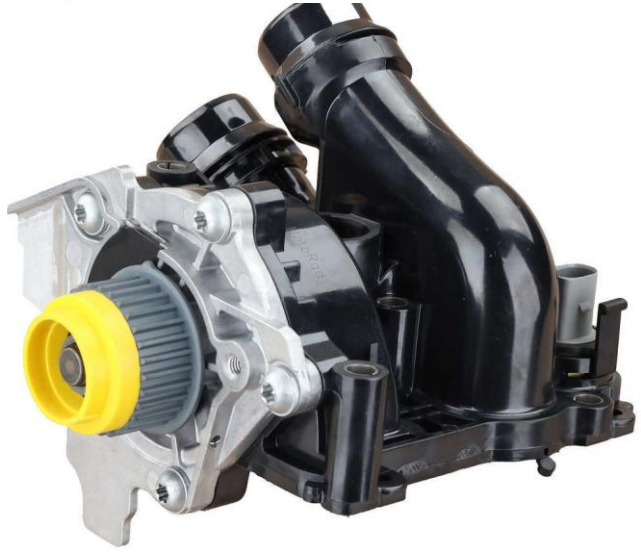
Types of water pump

- Mechanical
- Switchable
- Electric
- Auxiliary













The mechanical water pump

What drives it

– Mounted on the engine or built into the engine block, depending on the design
– Driven by the timing belt, the accessory belt or directly by the engine
– Its flow follows engine speed: low at idle, high at high rpm



GOOD TO KNOW — Pump driven by the timing belt: also replace the belt, the tensioner and the pulleys.

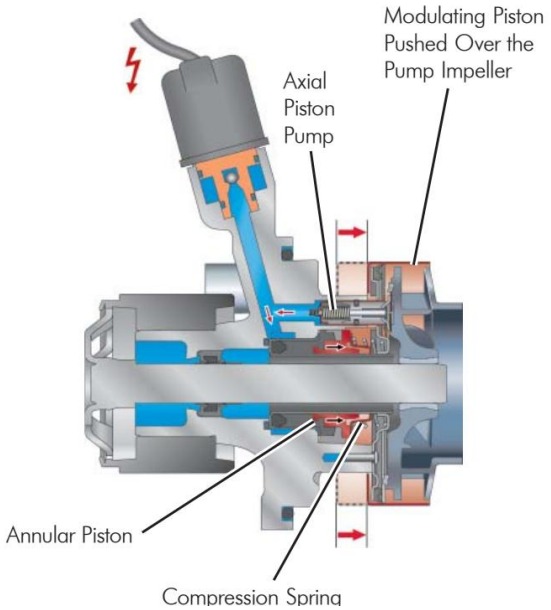
The switchable water pump

What it does

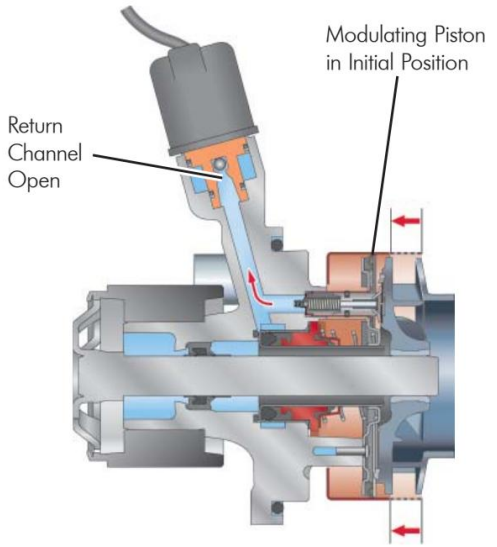
– A mechanical, belt-driven pump that can be stopped from delivering
– When cold, a shroud slides over the impeller: the coolant stays still and the engine warms up faster

The VW example: 2.0 TDI EA288

– The ECM controls valve N489: a small built-in hydraulic pump pushes the shroud over the impeller
– The shroud retracts according to cylinder head temperature, load and engine speed



Valve N489 activated: no flow



Valve N489 at rest: coolant flows

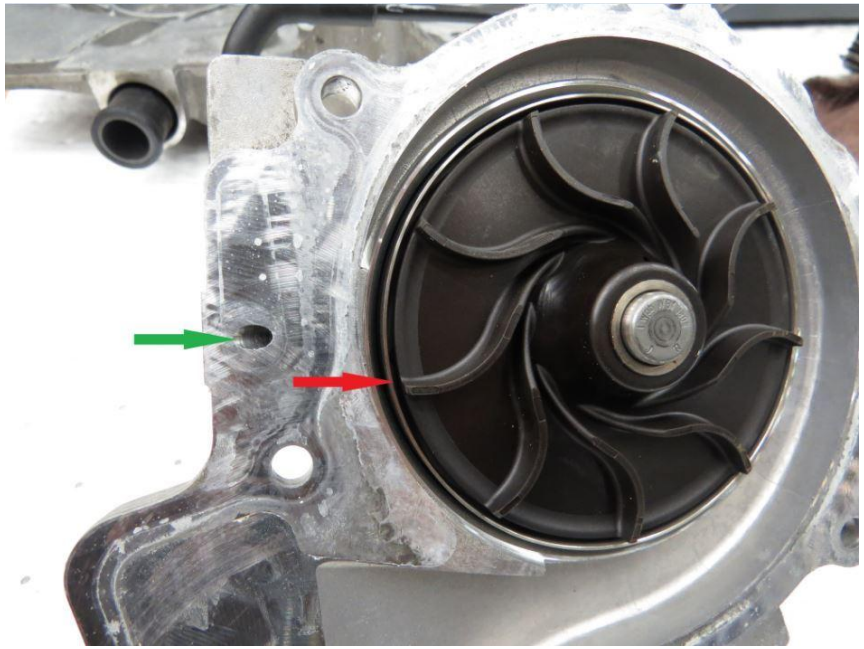
GOOD TO KNOW — If valve N489 fails, the shroud stays open and coolant flows: the engine does not overheat, it only warms up more slowly.

The vacuum-switched water pump

The Mercedes example: the Sprinter's OM651 diesel

How it works

– A switchover valve sends vacuum to a diaphragm
– The diaphragm slides a shroud over the impeller: flow is cut, the engine warms up faster
– With no vacuum, the shroud retracts and the pump delivers normally
– The vacuum comes from the same reservoir as the turbo, the EGR and the charge air flap



Green: vacuum port. Red: the shroud. (Mercedes)

GOOD TO KNOW — Same idea as the VW: a shroud over the impeller. Here, a vacuum leak can affect the pump and the turbo at the same time.

Sprinter OM651: coolant in the vacuum

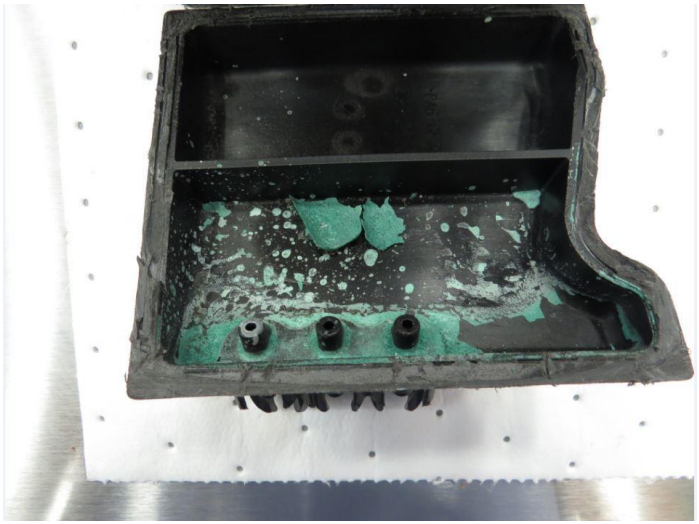
The real case: Mercedes bulletin LI07.09-N-063338

The symptoms

– Check engine light, limp mode, lack of power
– Engine that does not reach temperature (P012800) or runs hot
– Boost codes (P0299FA) and circulation pump codes (P261F71)

What Mercedes asks for

– Smoke test or pressure test, 1 bar maximum
– Disconnect the vacuum reservoir hoses and look for coolant or oil
– Coolant: replace the water pump, the reservoir and the switchover valve, clean the hoses
– Oil: replace the vacuum pump, the reservoir and the brake booster



Vacuum reservoir full of coolant



Water pump valve, contaminated

GOOD TO KNOW — The vacuum reservoir is sealed: do not open it, replace it. The photos show parts cut open for illustration.

Testing the vacuum-switched pump

Before replacing it

In order

1. The scan tool guided tests first (Mercedes: XENTRY)
2. Disconnect the vacuum hoses from the valve and the reservoir: coolant points to the pump diaphragm
3. Hand vacuum pump on the pump valve: the vacuum must hold
4. Smoke test or pressure test of the vacuum system, 1 bar maximum

Reading the symptom

– No vacuum: the shroud stays open, the engine warms up slowly (P012800)
– Ruptured diaphragm: coolant enters the vacuum system, then the turbo and EGR follow

GOOD TO KNOW — Never blow out vacuum hoses with compressed air: it damages the other components. Let them dry.

The electric water pump

What it does

– An electric motor drives the impeller: flow no longer depends on engine speed
– The computer controls it according to actual need: less friction, fuel consumption and emissions



The BMW example (turbo engines, including the N55)

– Shut off completely during warm-up
– It keeps running with the engine off, if it is very hot
– The DME targets 108 °C in economy driving, and down to 90 °C at full load



GOOD TO KNOW — An electric pump that runs after the engine is shut off is not faulty: it is the after-run.

The auxiliary water pump

What it does

– A small electric pump that assists the main pump or serves a separate circuit
– Controlled by the ECM, often with a PWM signal



Where you find it

– VW/Audi: pump V51, which boosts heating and cools the turbo after shutdown
– Liquid-to-air charge air cooler (VW: pump V188)
– Hybrids and EVs: battery, power electronics, electric motor



GOOD TO KNOW — On a recent VW or Audi, weak heat can come from the auxiliary pump, not only from the thermostat.

The electric pump: what to check

Before replacing it

In order

1. Read the codes: pump circuit, speed or blockage
2. Command it with the scan tool: you should hear it and see coolant move in the reservoir
3. At the connector: power, ground and control signal (PWM or network)

The traps

– Never run it dry: the bearing gets damaged
– During a forced activation, keep the 12 V battery on a charger

GOOD TO KNOW — A pump that is noisy and makes bubbles in the reservoir during activation: it is often air in the system, not the pump.

The weep hole

What the weep hole tells you

What it is for

– A deliberate opening, between the seal and the bearing
– It drains the coolant that gets past the seal, so it does not flood the bearing

Normal or leak?

– Dry trace or slight dampness, especially after installation: normal, the coolant lubricates the seal
– Running drip, puddle, dropping level: the seal is finished, replace the pump

The documented causes

– Contaminated or incompatible coolant, debris, mineral deposits, cavitation



Suspended debris



Incompatible coolant



Cavitation



Leak due to contamination

GOOD TO KNOW — Before condemning a pump: clean, pressure test, then check again. A dry trace is not a leak.

The bearing and the shaft

Three checks, engine off

Play

– Grab the pulley or the fan and try to move it in every direction, no play should be felt
– Any play comes from the pump bearing or from the fan clutch bearing

Noise

– A rumble that follows engine speed and disappears with the belt removed

The causes of failure

– Improper belt tension, which creates a high load
– Misalignment, which creates a side load
– A failing fan clutch, which creates excessive vibration
– Bent or incorrect blades, which create an imbalance

GOOD TO KNOW — Tired fan clutch: replace it with the pump, otherwise its vibrations destroy the new bearing.

The impeller and the housing

What to look at on a pump that has been removed

The impeller

– Plastic impeller eroded or eaten away: cavitation, or contaminated coolant
– Impeller loose on the shaft: it spins without pumping, the engine overheats for no apparent reason
– If the pump looks like this, flush the system before fitting the new one

The housing

– Cracked housing: overtightening, tightening sequence not followed, material trapped between the surfaces
– Surfaces not cleaned, uneven, or too much sealant: the leak comes back

GOOD TO KNOW — Housing cracked after installation: look at the tightening first, to the manufacturer’s torque and sequence.

Fitting the right pump

Three traps at replacement

The rear housing

– Many pumps are sold with the rear housing preassembled: one part number, one gasket, less risk of a leak

The pulley

– The pin spanner exists to hold the pulley without scratching or binding it when removing the fan clutch

The hybrid

– Hybrid vehicle pumps have their own part number, never assume a pump from the same displacement will fit

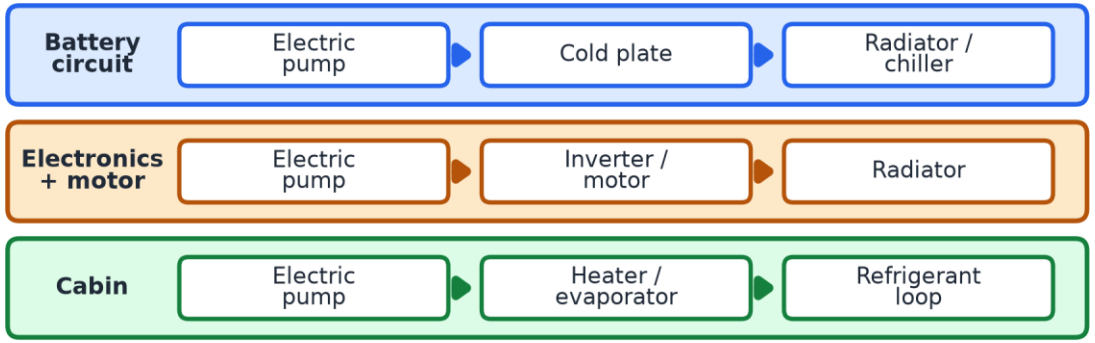
WARNING — Bleed the system properly after fitting. An air pocket makes a new pump cavitate and destroys it within a few weeks.

EV cooling

Several circuits, several pumps

How it is organized

– An EV has several separate glycol loops, each with its own electric pump or pumps
– The glycol carries and redistributes the heat; valves link the circuits together



Valves link the circuits: series or parallel, and heat sharing.

GOOD TO KNOW — The refrigerant side (heat pump, chiller, A/C) is covered in detail in the Heat Pump training.

Auxiliary pumps

The heart of circulation on an EV

Why they are central

– With no engine belt, all circulation depends on electric pumps; an EV has several, one per circuit plus dedicated pumps
– Controlled by the thermal management module, they match flow to demand and run even with the vehicle parked, while charging

Diagnosis

– A failing pump makes a single component overheat, battery or inverter, without classic engine overheating
– Read the pump codes, command it with the scan tool, then check power and signal

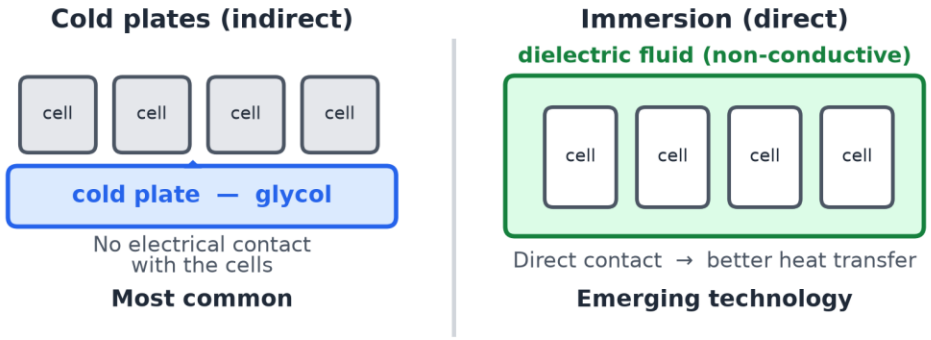
GOOD TO KNOW — Never open the reservoir when the high-voltage system is hot: the coolant is under pressure and burns.

Cooling the battery

Two main methods

The two approaches

– Cold plates: the glycol flows under the cells, with no electrical contact. The most common
– Immersion: cells in a dielectric fluid. Better transfer, emerging technology



GOOD TO KNOW — With cold plates, the coolant does not touch the cells, but it runs right next to the high voltage: hence low conductivity.

Conductivity values

Service and diagnosis

What you do in the shop

– Deionizer (ion exchange resin): for now only on FCEVs, for example the Toyota Mirai and the Hyundai Nexo
– On battery EVs: measure with a conductivity meter and replace the coolant above the manufacturer’s limit



GOOD TO KNOW — The limit to respect is the manufacturer’s. The standard’s 100 µS/cm threshold applies to new coolant.

A real case: draining an Ioniq 5

Battery circuit, low-conductivity coolant

Drain and fill

– Battery and radiator cool to the touch before opening the reservoir
– Drain through the battery electric pump hose, under 0.2 MPa of air, with the Hyundai tool
– Vacuum filling with the same tool (vac-n-fill)

Bleed and check

– “EWP operation for degassing” mode on the scan tool, 12 V battery on a charger
– Top up to MAX when there are no more bubbles or noise
– Recheck the level for 2 to 3 days

GOOD TO KNOW — Low conductivity: never water, never additives, never two brands mixed. About 9 to 12 L depending on the version.

Why active thermal management?

What it aims for

– Warm the engine up as fast as possible: less friction, catalyst effective sooner, faster heat
– Set the temperature according to load, rather than a fixed value
– Also cool the oil, the transmission and the turbo, even after shutdown



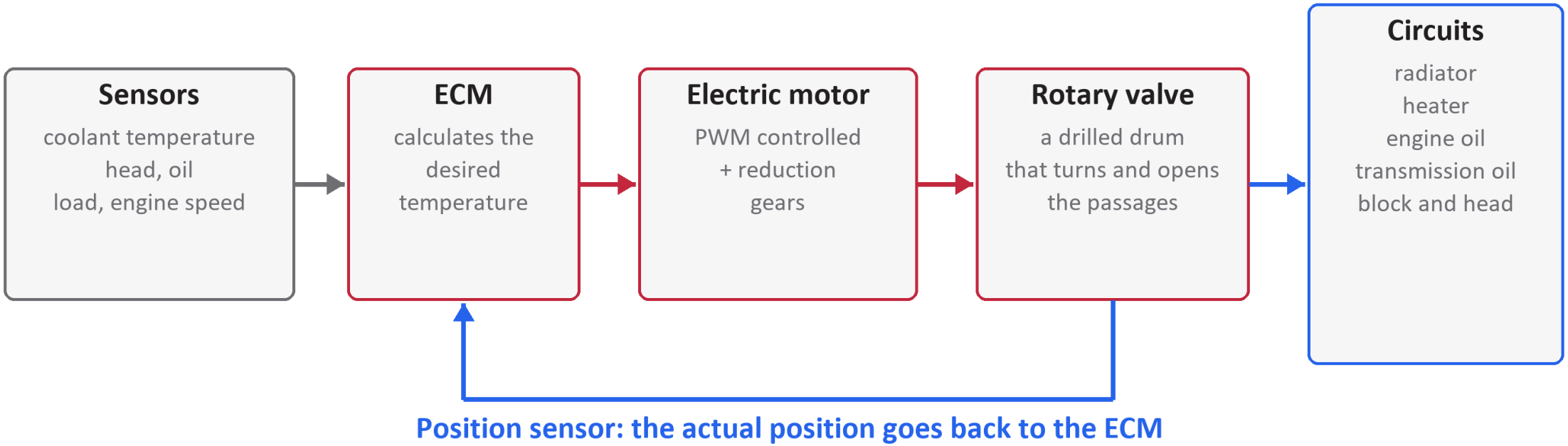
Thermal management modules from several manufacturers

VW example (EA888)

– 107 °C at light load, 85 °C at heavy load

GOOD TO KNOW — The wax thermostat reacts to coolant temperature. Active thermal management anticipates: the ECM decides based on load, engine speed and outside temperature.

The common principle



GOOD TO KNOW — The position signal varies by manufacturer: 0 to 5 V voltage (Mazda, Hyundai), SENT digital signal (VW, BMW) or LIN network. The ECM constantly compares the target with the actual: it is the gap that sets the codes.

The strategies

Five ways to manage heat

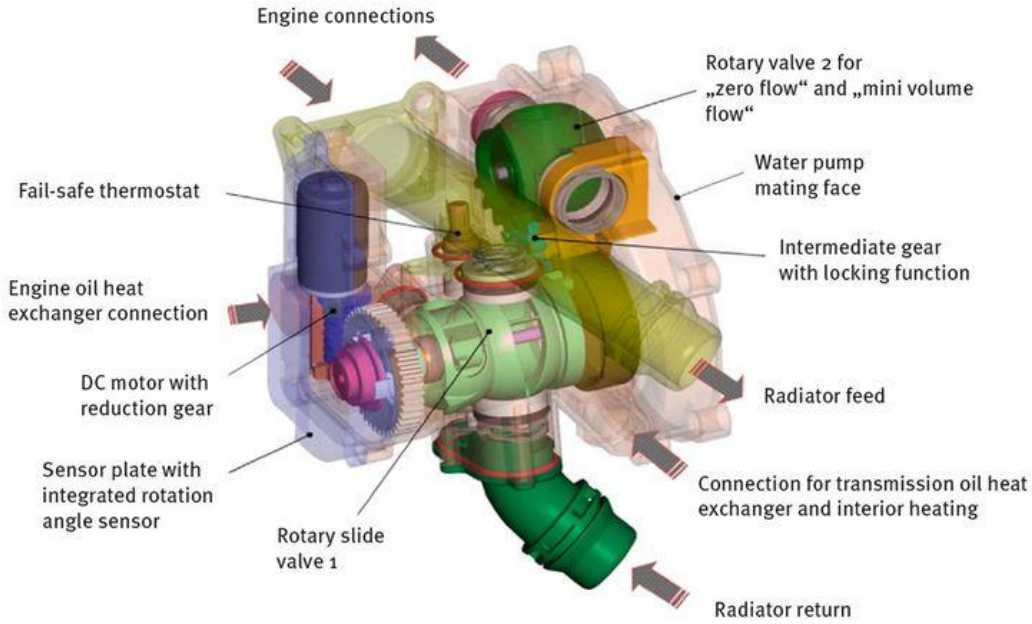
– Zero flow at start-up: coolant stays still in the block (VW: up to 90 °C)
– Micro-flow: a trickle of coolant protects the cylinder head and the turbo
– Separate head and block: the head is cooled first, the block kept hotter
– After-run: after shutdown, an electric pump keeps circulating (VW: up to 15 min)
– In case of failure: wax backup thermostat or fail-safe position, and limited power

GOOD TO KNOW — A cold hose on a hot engine is no longer necessarily a fault: the ECM may have closed that circuit on purpose.

VW/Audi EA888: the thermal module

What it contains

– The water pump, always belt-driven
– Two rotary slide valves: no. 1 handles radiator, oil and head; no. 2, the block
– Actuator N493: DC motor, worm gear, angle sensor
– A wax backup thermostat



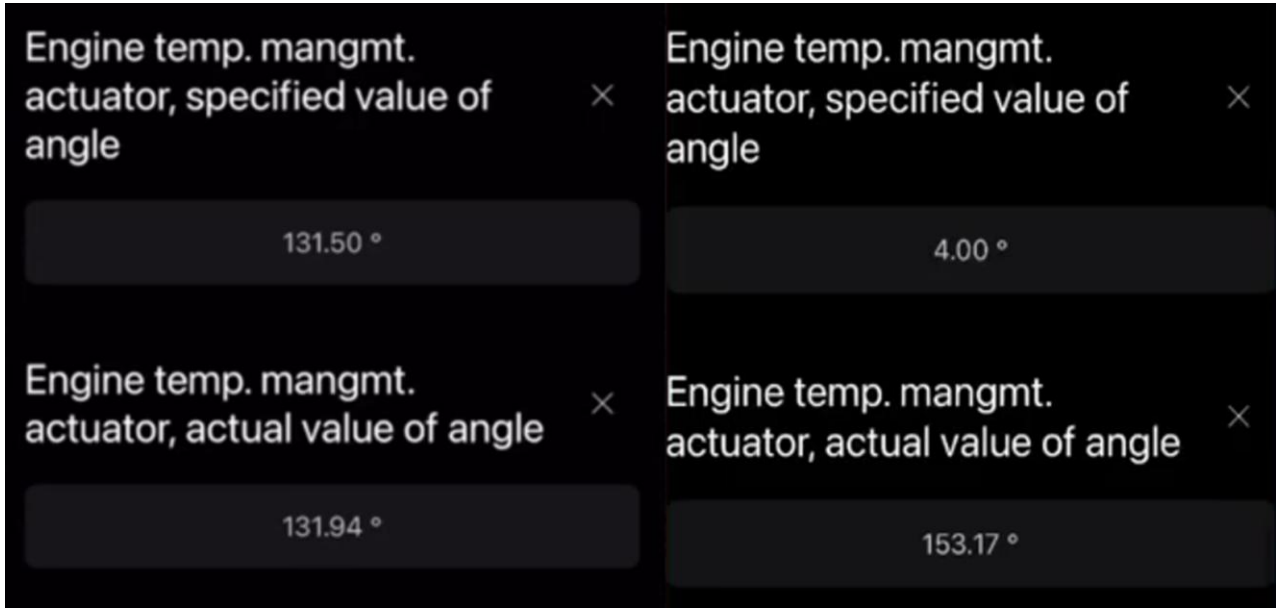
GOOD TO KNOW — 1.8 and 2.0 TSI EA888 3rd generation engines, from about 2015: Golf, GTI, Jetta, Passat, Tiguan, Audi A3, A4, Q5. It is not a variable pump: the pump is mechanical, the slide valves meter the flow.

VW/Audi EA888: the warm-up

Phase	Slide valve 1	What happens
Cold start	160°	Coolant still in the block, up to 90 °C maximum
Micro-flow	Around 145°	Slide valve 2 opens slightly: a trickle of coolant protects head and turbo
Oil	From 120°	The oil cooler comes into play: the oil warms up faster
Regulation	From 85° to 0°	The radiator opens: 107 °C at light load, 85 °C at heavy load
After shutdown	From 160° to 255°	After-run up to 15 min, with the electric pump

> **GOOD TO KNOW —** On the scan tool, the angle of rotary slide valve 1 is read in degrees: it tells which phase the engine is in.

VW/Audi EA888: when it fails



Good: target 131.5°, actual 131.9°

Stuck: target 4.0°, actual 153.2°



Coolant has reached the motor and the gears

The signs and the limp mode

– Code P00B7, insufficient coolant flow (recorded on a 2016 VW, 1.8 TSI)
– Angle sensor failed: maximum opening. Motor or slide valve stuck: engine speed limited to 4,000 rpm, EPC light
– Above 113 °C, the wax capsule opens toward the radiator

GM 3.0 L Duramax: two circuits

High temperature

– Mechanical pump: constant flow to the block, head and valves
– A three-valve housing: flow control, main valve, block valve
– Engine and transmission oil heat exchangers heat or cool

Low temperature

– Electric pump on the frame and auxiliary radiator
– Charge air cooler (liquid-to-air)
– Fuel cooler and DEF injectors



Three-valve housing



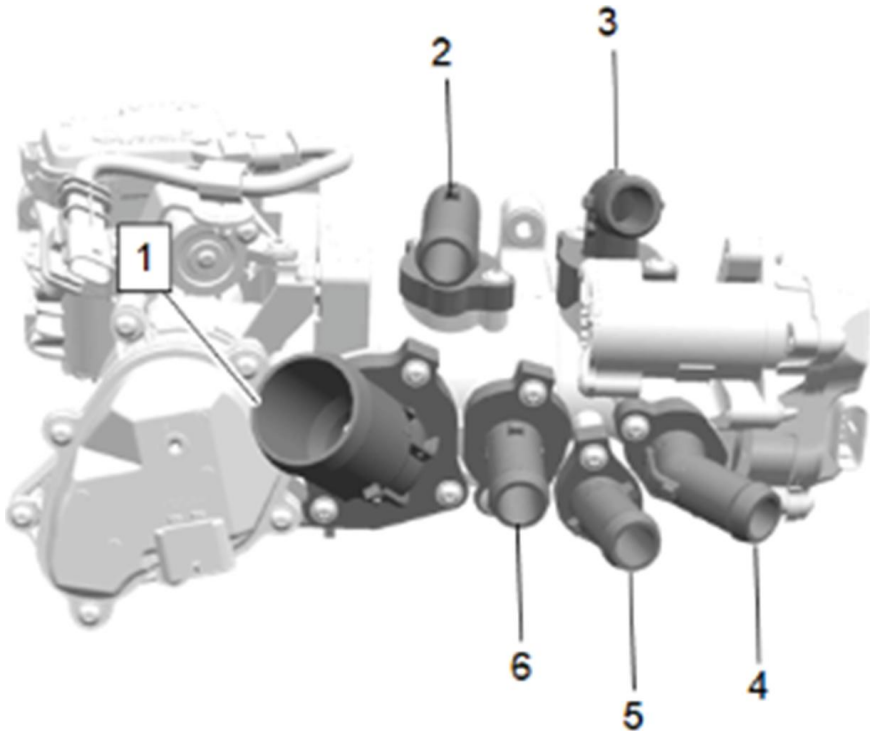
Low-temperature electric pump

GOOD TO KNOW — LM2 engine: Silverado and Sierra 1500, Tahoe, Suburban, Yukon, Escalade. The valves replace the wax thermostat.

GM 3.0 L Duramax: the six ports

The flow control valve

– 1. Large side port: flow to the radiator
– 2. Top port, always open: EGR, turbo and heater
– 3. Right side port: EGR and turbo return
– 4. Lower right port: cold coolant from the heater pump
– 5. Small valve to the left of 4: to the oil heat exchangers
– 6. Bypass port



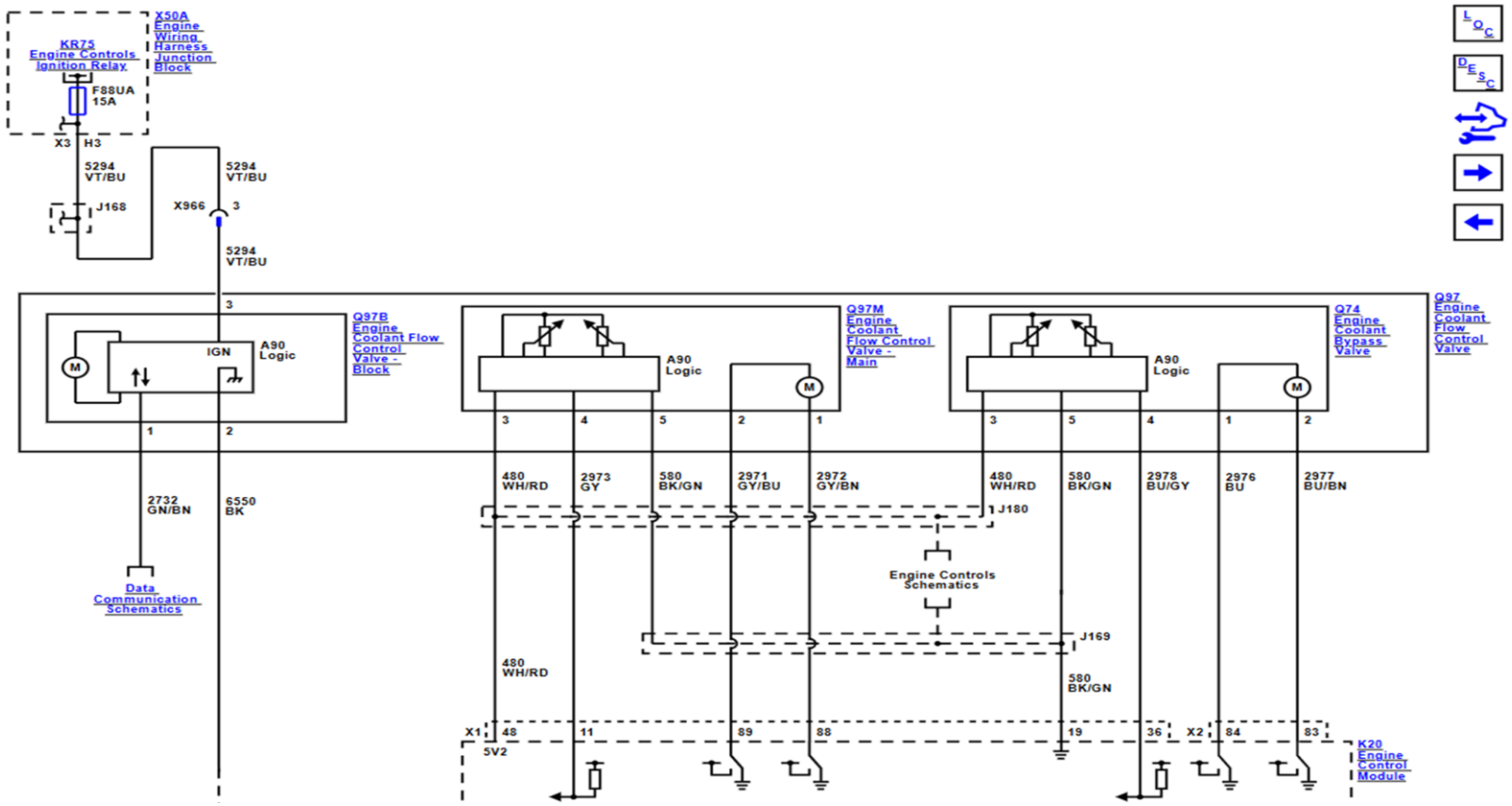
GOOD TO KNOW — The ECM decides, based on about a dozen sensors: coolant, head, oil and transmission temperatures, and the position of each valve.

GM 3.0 L Duramax: the seven modes

Mode	What happens
1. Zero flow	Cold engine: nothing circulates, heat stays around the combustion chamber
2. Cabin heating only	Coolant from the head, EGR and turbo goes to the heater; radiator closed
3. Heating with bypass	Coolant circulates in the engine without passing through the radiator
4. Heating and oil	Engine oil and transmission oil are also warmed
5. Oil heating	Hot engine: the radiator opens as needed
6. Oil cooling	Cooled coolant goes to the oil heat exchangers
7. After shutdown	Auxiliary pump at low speed, main valve at maximum, block valve open

GOOD TO KNOW — Zero flow is not enabled if heat is requested. At shutdown, the valves stay open and run a self-test that may be audible: it is not a fault.

GM 3.0 L Duramax: the wiring

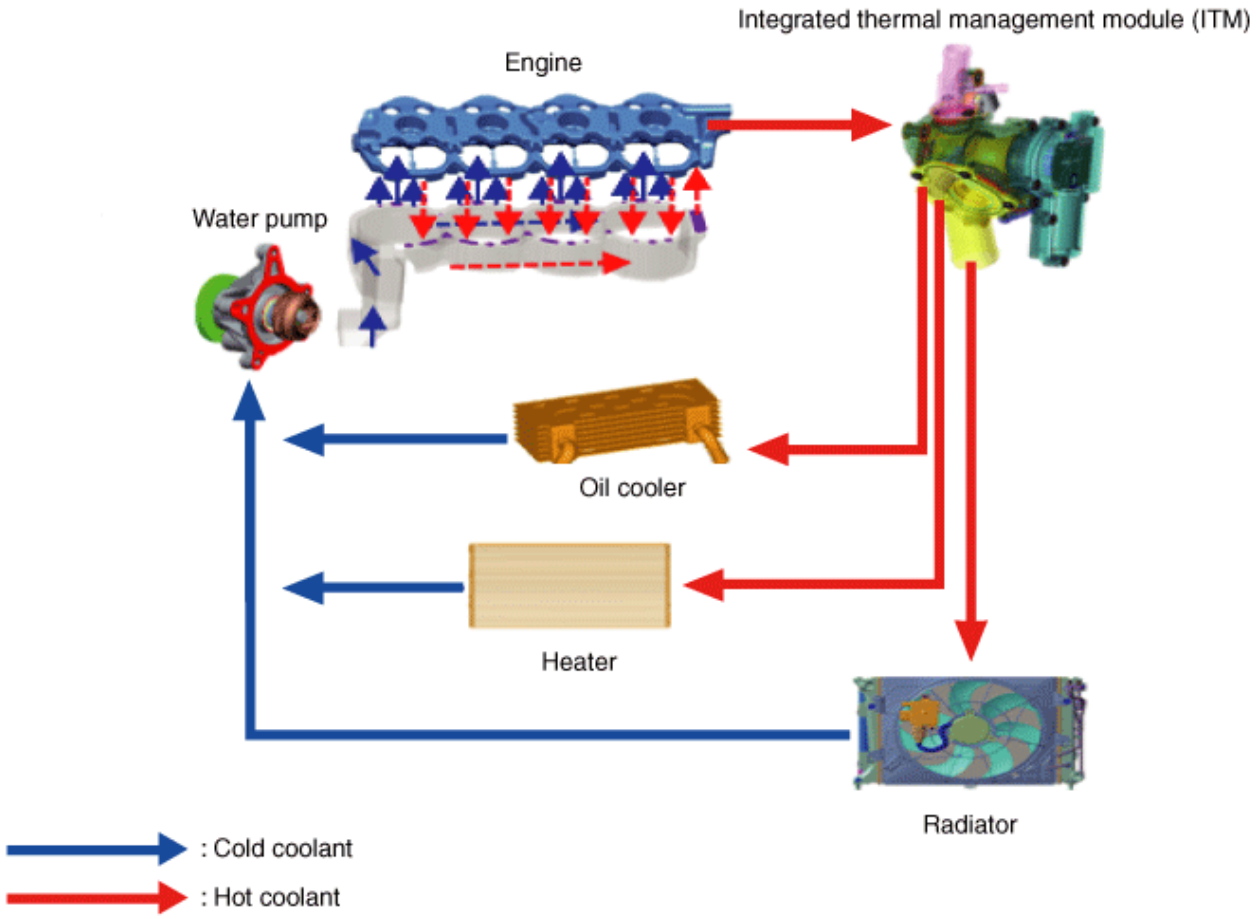


Three actuators

– Block valve (Q97B): 12 V, ground and data network
– Main valve (Q97M) and bypass valve (Q74): a two-wire motor and a three-wire position sensor (5 V, signal, ground)

GOOD TO KNOW — Before condemning a valve: check the 15 A fuse, the ground and the 5 V reference at the connector.

Hyundai: the ITM module



What it does

– A single housing directs coolant to the radiator, the heater and the oil cooler

Where it is found

– 2.5 L and 2.5 T Smartstream engines: Sonata, Santa Fe, Tucson (2020 to 2023)

GOOD TO KNOW — ITM: Integrated Thermal Management, Hyundai's name for its thermal management module.

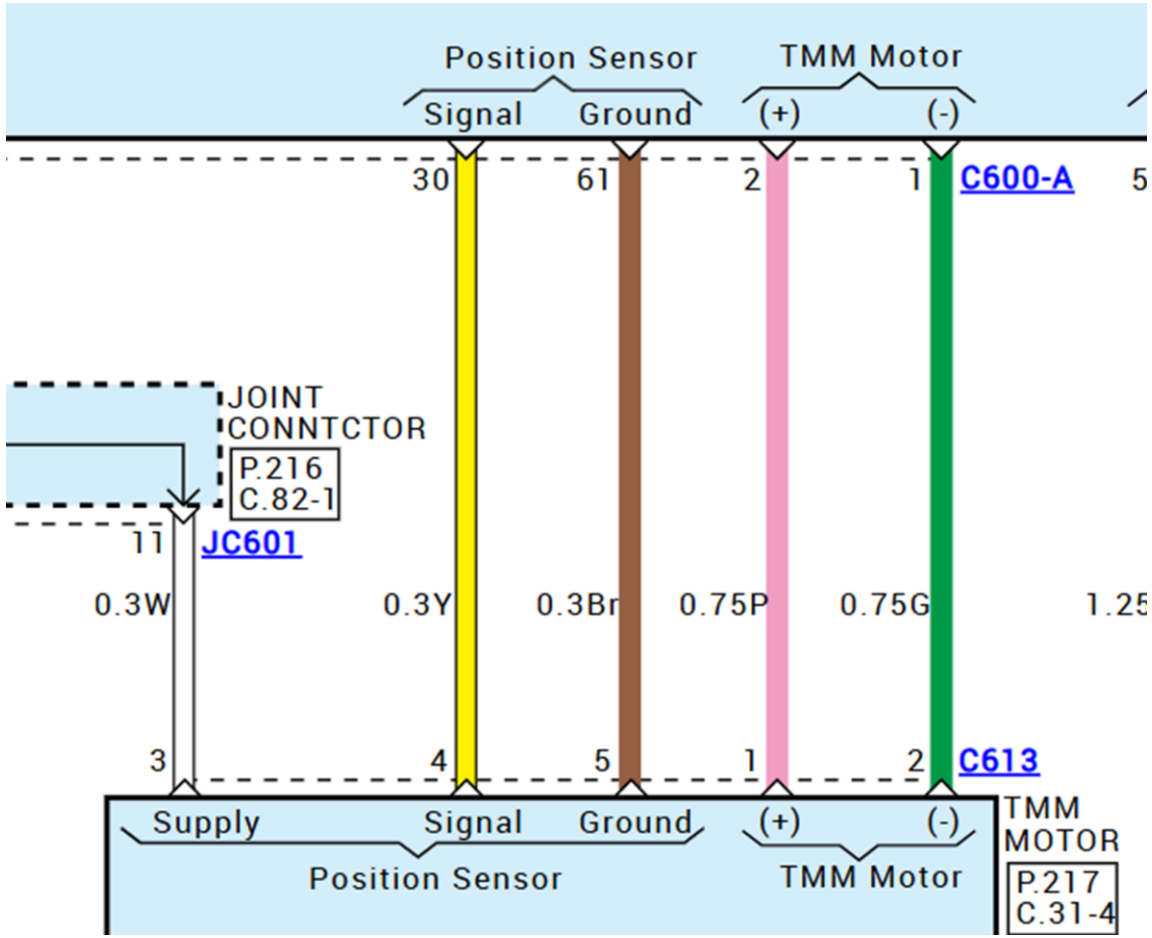
Hyundai: the wiring and P2181

The connector

– Position sensor: 5 V, signal, ground
– Motor: two wires, driven by the ECM from 0 to 12 V

The real case: bulletin 24-EM-001H

– Code P2181, cooling system performance
– Replace the ITM with the revised part
– Bleed with the ITM filling mode of the GDS scan tool



GOOD TO KNOW — Without the filling mode on the scan tool, air stays trapped in the system. We come back to this with draining.

Mazda: the coolant control valve

What it is

– SKYACTIV-G 2.0 and 2.5 L: a motorized valve with a built-in backup thermostat, controlled by the PCM

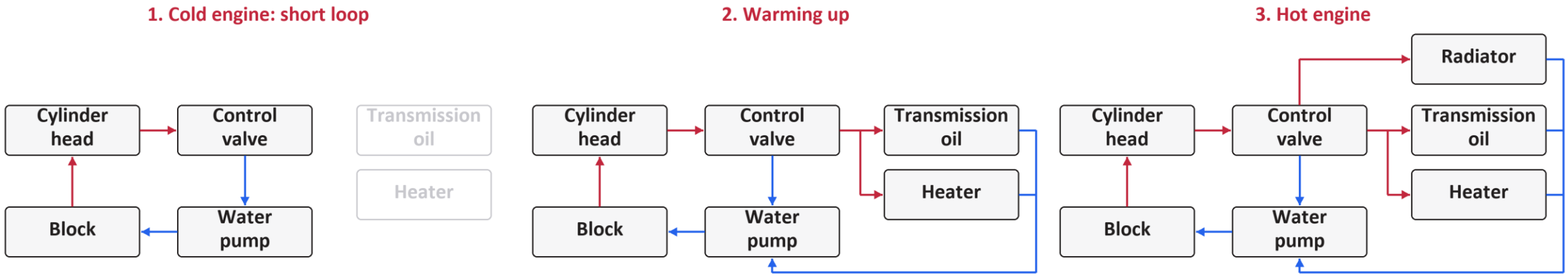
The real case: bulletin 01-011/21

– Code P0126 (thermostat stuck open), engine slow to warm up, weak heat at idle
– Cause: the backup thermostat opens too early, between 40 and 80 °C
– Fix: replace the valve (PYFD-15-16Z)



GOOD TO KNOW — Mazda3 2019-2021, Mazda6 2018-2021, CX-3 2019-2021, CX-5 2018-2021, CX-30 2020-2021, except 2.5 L turbo. The test: read the coolant temperature in the code's freeze frame.

Mazda: the circuit by temperature



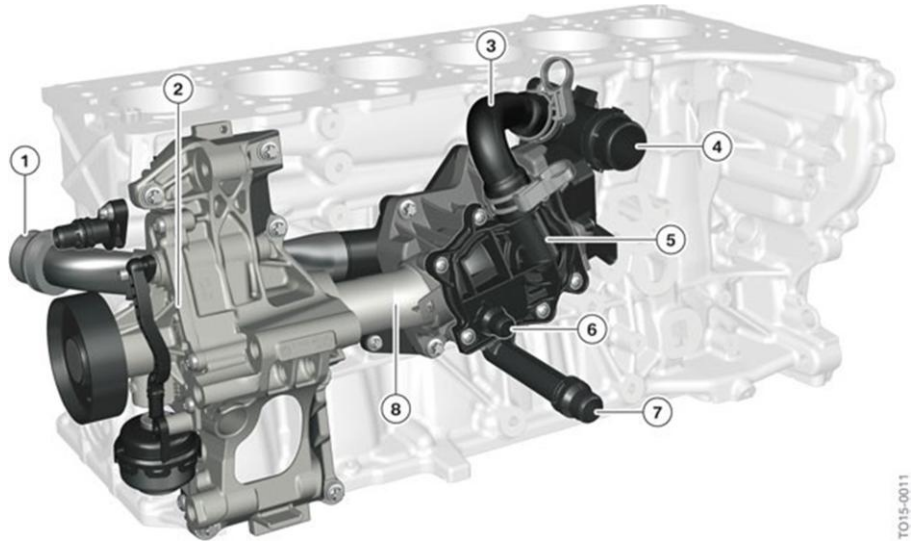
The wiring

– Valve motor: two wires driven by the PCM
– Position sensor: 5 V, signal, ground

GOOD TO KNOW — Cold engine, coolant circulates in a short loop without the radiator. The radiator only opens at the end, when the engine is hot.

BMW: the thermal management module

Command	Bypass	Heater	Radiator
0 %	Closed	Closed	Closed
5 %	Partly open	Closed	Closed
6 %	Partly open	Partly open	Closed
10 %	Open	Open	Closed
11 %	Open	Open	Partly open
90 %	Almost closed	Open	Open
100 %	Closed	Almost closed	Open



The wiring

– Motor: two wires
– Sensor: 5 V, SENT digital signal, ground

GOOD TO KNOW — The DME commands the module as a percentage. At 0%, nothing circulates: it is the zero flow of a cold start.

Diagnosing active thermal management

Four steps, in order

– 1. Read the code: circuit (electrical), position or performance (mechanical), communication (network)
– 2. Compare the target and the actual valve position on the scan tool
– 3. Run an actuator test with the scan tool and see if the position follows
– 4. At the connector: power, ground, 5 V reference and position signal
– Never apply 12 V directly to an actuator to test it: use the scan tool

GOOD TO KNOW — A gap between target and actual with a healthy electrical circuit is often mechanical: broken gear, drum stuck by deposits or swollen seals.

After replacement

Three steps not to skip

– Vacuum fill: air stays trapped in the module and the head
– Bleed with the scan tool or vehicle function
– Run the end-stop learning if the manufacturer requires it

Examples

– Hyundai: ITM filling mode of the GDS scan tool
– GM Duramax: automatic fill, started with the scan tool or with the driver controls
– VW, Honda, GM: bleeding procedures covered at the end of the training

GOOD TO KNOW — A new part that sets a position code, or an engine that overheats on first start-up: it is almost always a missed bleed or learning procedure.

Coolant, the module’s number one enemy

What damages it

– A non-compliant coolant: the seals swell and the valve sticks
– Oil in the coolant (leaking oil cooler): same effect
– Coolant that gets past the seals and reaches the motor and the gears
– Deposits from worn-out or diluted coolant



GOOD TO KNOW — Before installing a new module: drain, flush, and fill with coolant that meets the manufacturer’s specification. Otherwise, the new one goes the same way.

Overheating, the full sweep

The possible causes, in the order you check them

Airflow
Fan not running when it should, failing clutch, broken blades, obstructed airflow

Radiator
Internal blockage, contamination, broken fins

Lower hose
Collapse under pump suction

Thermostat
Stuck closed, installed backwards

Cap
Improper working pressure, wrong rating, boiling point too low

Coolant
Wrong mixture, level too low, trapped air, improper bleeding

Drive
Tensioner, incorrect belt routing

Engine
Restricted exhaust, mixture too lean, head gasket, ignition timing

GOOD TO KNOW — Belt routed wrong after a replacement: the pump can turn backwards and the engine overheats.

The operating condition guides the diagnosis

When it overheats tells you a lot

It overheats at idle or low speed, and settles down while driving

– Air is what is missing: fan, fan clutch, missing deflectors, radiator plugged on the air side

It overheats on the road, under load, but not at idle

– Coolant is what is missing or flowing poorly: pump, thermostat, radiator plugged inside, low level

It overheats all the time, right away

– Little or no circulation: closed thermostat, detached impeller, large air pocket

WARNING — A needle that swings up and down sharply is almost always a low level or air, not a thermostat.

When the engine protects itself

Complaints that do not look like overheating

BMW: two stages

– Coolant at 117 to 124 °C or oil at 150 to 157 °C: the A/C is reduced, up to shutting it off, then power
– Coolant at 125 to 129 °C or oil at 158 to 163 °C: power reduced by up to about 90%

VW/Audi EA888: thermal module failure

– Engine speed limited to 4,000 rpm, EPC light, warning chime and temperature displayed
– The module goes to maximum cooling to protect the engine

GOOD TO KNOW — A/C cutting out or loss of power in hot weather: look at the coolant and oil temperatures on the scan tool.

No heat on a recent vehicle

What comes on top of the classic causes

What can cut off the heat

– A controlled heater valve stuck closed (VW/Audi: N82, N422)
– A failed heater pump (VW/Audi: V51 or V50)
– A thermal management valve that opens too early: Mazda, P0126, weak heat at idle

What is normal

– When cold, the computer may keep the heater closed to warm the engine first
– A heat request cancels zero flow (GM Duramax) or opens the valve (VW), at the cost of a slower warm-up

GOOD TO KNOW — Before replacing a heater core: check on the scan tool whether the heater valve and pump are being commanded.

What costs the most afterwards

Three recurring mistakes

Not bleeding the system

– An air pocket makes the new pump cavitate, creates local heating and destroys what was just installed

Not calibrating

– On active thermal management systems, a valve or pump replaced without learning on the scan tool gives a position code and incorrect behaviour

Using tap water

– Minerals, scale, corrosion. It is a roadside fix, not a repair

Draining and vacuum filling

Why the vacuum tool

The rules

– Never put the old coolant back in
– Vacuum fill: the coolant gets everywhere, with no air pocket
– Follow the manufacturer’s procedure: some require the vacuum tool (Hyundai Ioniq 5)





GOOD TO KNOW — Vacuum does not replace bleeding: on an engine with thermal management, you still need the scan tool or vehicle function.

Draining the system

The procedure

The steps

– Drain the system through the radiator drain if possible, or through a hose, as the manufacturer recommends
– Fill with the coolant required by the manufacturer, to the specification
– Bleed the entire system according to the manufacturer
– Recheck the level cold, the next day

KEY POINT — Bleeding follows the manufacturer’s method: with the scan tool, with a vehicle command or at engine speed.

GM: Auto Coolant Service Fill (ACSF)

A bleed commanded by the vehicle

What the technician does

– Start ACSF: accelerator and brake pressed for 2 s, then release. No message in the cluster, up to 60 s before it becomes audible
– 1,200 rpm for 3 min while topping up the surge tank, then install the cap
– Alternate 3,500 and 1,200 rpm, 3 min each, while watching the temperature
– Let it cool, adjust to the cold fill line, check for leaks

7. Press the accelerator and brake pedal for two seconds at the same time, then release.
   Note:
   - There will NOT be a message on the Driver Information Center (DIC) to indicate that the ACSF is activated.
   - It may take up to 60 seconds for the ACSF procedure to become audible.
8. Start and run the engine at 1200 RPM for three minutes while adding coolant mixture as the level drops below the fill mark on the radiator surge tank until the level stabilizes.
9. Install the radiator surge tank cap.
10. Run the engine as follows while watching the coolant temperature gauge ensuring that the temperature is rising, but not overheating. If the radiator surge tank empties during the procedure below, stop the process and fill the radiator surge tank with the coolant mixture to the fill mark and start the procedure over.
    1. Run the engine speed at 3500 RPM for three minutes.
    2. Lower the engine speed to 1200 RPM and run for three minutes.
    3. Raise the engine speed to 3500 RPM and run for three minutes.
    4. Lower the engine speed to 1200 RPM and run for three minutes.

    Note: If you are unable to maintain the coolant level at the cold fill line of the surge tank and the surge tank empties while the process is running, turn the ignition off. The vehicle will only allow one ACSF process per key cycle.

    1. Fill the radiator surge tank to the cold fill line and wait two minutes.
    2. Repeat steps 4-10 until the ACSF cycle completes with the coolant level maintained in the surge tank for the entire process.
11. Turn ignition off.
12. Allow the engine to cool
13. Adjust the coolant level in the surge tank to the cold fill line.
14. Install the radiator surge tank cap.
15. Start the engine and check for coolant leaks.

GM service information, 3.0 L Duramax LM2, steps 7 to 15

GOOD TO KNOW — Only one ACSF cycle per key cycle: if the tank empties, turn the ignition off, fill to the cold line, wait 2 min and start over.

VW and Honda: two ways to bleed

The method changes from one manufacturer to another

VW/Audi: with the tool

– Auxiliary heater, if fitted, switched on for about 30 s

– Connect the diagnostic tester, start “01 Coolant Circuit Bleeding” and follow the screen

Honda: at engine speed

– Engine slightly above idle until the cooling fan comes on twice, A/C off

– Stop, check the level in the tank, top up with Honda Type 2

Vehicles on which the bleeding procedure is stored in the engine control module:

– If the vehicle has a parking/auxiliary heater, switch it on for about 30 seconds.

– Connect the Vehicle Diagnostic Tester .

– Start the program 01 Coolant Circuit Bleeding using the → Vehicle Diagnostic Tester and follow the instructions in the vehicle diagnostic tester display.

8. Start the engine and warm up the engine by slightly raising the engine speed above its normal idle speed until the cooling fan(s) comes on twice. Do not operate the air conditioning system.

9. Turn off the engine. Check the level in the tank, and add Honda Long Life Antifreeze/Coolant Type 2, if needed.

GOOD TO KNOW — Bleeding is no longer universal: check the manufacturer’s method before starting, tool or engine speed.

Hyundai: ITM coolant filling mode

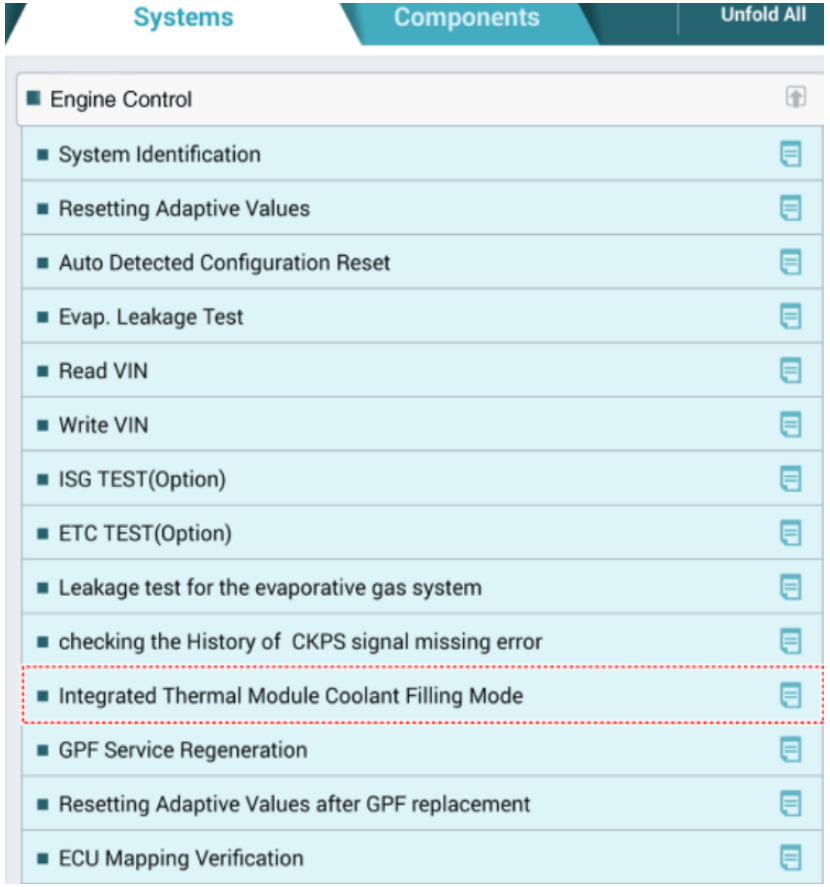
A bleed commanded by the GDS scan tool

Where to find it

– GDS: Engine Control, then “Integrated Thermal Module Coolant Filling Mode”
– Required by bulletin 24-EM-001H after replacing the ITM (code P2181)

Why

– Without this mode, air stays trapped in the system



GOOD TO KNOW — Same logic at VW, GM and Hyundai: a thermal management module is bled with electronics, not by hand.



Closing words !

Thank you all for taking part!

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